SPIDER radio telescope user manual

update 16/07/2019

This manual describes the use of the SPIDER radio telescope with RadioUniversePRO software. Before using the radio telescope, user has to check the radio telescope status and follow safety rules.

- SAFETY RULE: by using the SPIDER radio telescope, the user understand that no person has to stay close to the radio telescope since the antenna may move and hit the person if too close to it. The user accept PrimaLuceLab iSrl can't be held responsible if someone hurts or get injured being too close to the radio telescope. Especially in bad weather conditions no person is authorized to be close to the radio telescope that has to be controlled from the remote control room (where the receiver is located).

- This user manual is written by PrimaLuceLab iSrl and it's provided to SPIDER radio telescope owners. In any case the user is not sure on how to properly use the radio telescope, the user has to contact Radio2Space team (info@radio2space.com or +39/0434/1696106) BEFORE using the radio telescope.

Quick Start guide

In order to properly use the SPIDER radio telescope with the RadioUniversePRO software, please use this quick guide as reference and go to the next paragraphs to know more about the software features:

1) Turn on the mount and receiver power by pressing the power button in receiver and mount (mount box installed in the control room if you have SPIDER 300A or 500A, power ON button on the mount head if you have SPIDER 230C).
2) Start RadioUniversePRO software installed in your computer (please verify that the "Run as administration" option is selected. In order to do that, please go to RadioUniversePRO installation directory, for example c:/Programs/RadioUniversePRO, make a right-slick on RadioUniversePRO.exe file, select "Compatibility" tab and select "Run as Administration" option).
3) Connect to the mount by selecting the proper option in the left column.
4) Connect to the receiver by selecting the proper option in the right column.
5) Verify that the left column correctly shows the actual radio telescope position under "Actual Antenna position" (if you have SPIDER 300A or 500A, it should show the stow position AZ and EL coordinates that are Azimuth=0 and Elevation=90; if you have SPIDER 230C it should show Azimuth close to 0 and Elevation close to your Latitude if you start from mount HOME position).
6) Select Source Visibility tab and make a mouse double click on one of the visible (green colored) radio sources to move the radio telescope on the selected radio source.
7) Look at IF monitor and verify the presence of artificial interferences. If present, please use the BBC tool instrument.
8) Perform an automatic alignment of the radio telescope by using the Offset Alignment tool.
9) Record your data by using the proper OnOff, Total Power Plots and Mapping features.
10) When you finish using the radio telescope, park it in stow position (Azimuth=0 and Elevation=90).
11) Disconnect the mount and receiver in the RadioUniversePRO software.
12) Turn off the power from the receiver box and mount.

Safety notes

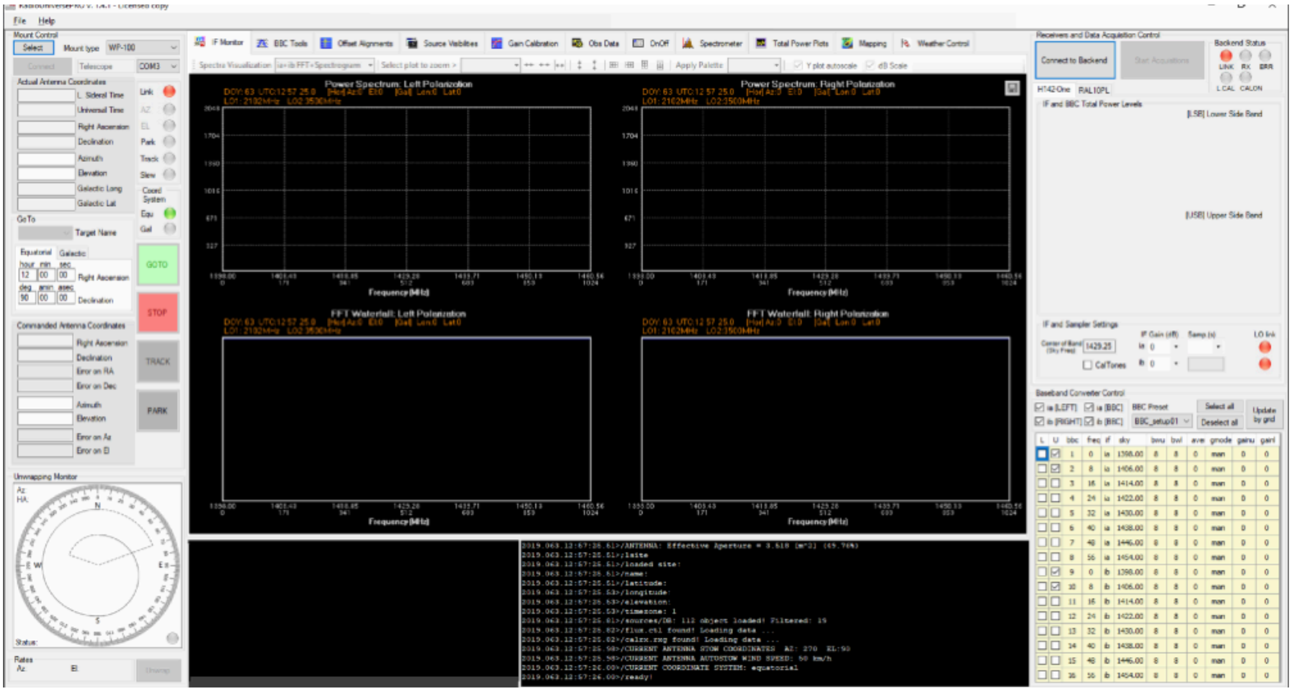
- SPIDER 300A and SPIDER 500A radio telescopes are designed to be operational with winds up to 50 km/h). If winds are higher than this value, please don't use the radio telescope and keep it in stow position (antenna pointed to the Zenith). If you are going to experience very strong winds, you can lock the antenna on the pier by using the provided tool (only for SPIDER 300A) and please remember to remove it before using the radio telescope again.

- SPIDER 230C radio telescopes is designed to be operational with winds up to 25 Km/hour - 15 miles per hour). If winds are higher than this value, please don't use the radio telescope and keep it in stow position (antenna pointed to the Zenith).

- RadioUniversePRO software do not allow the radio telescope to point object too close to the horizon, this value is 15 degrees by default. But if you want (and at your risk) you can reduce this value. In order to do this, just exit from RadioUniversePRO software go to RadioUniversePRO installation directory (it should be C:/program files/RadioUniversePRO) and open the file my settings.ini with Windows Notepad. Find the "elevation" value under mount settings (you will see it's 15) and reduce to 10. Then Save and exit from Note. Now you can start RadioUniversePRO again.

First use of RadioUniversePRO software

Please double clicking on the RadioUniversePRO icon in your computer desktop, this will open RadioUniversePRO software. RadioUniversePRO software is designed to allow you to have, also in a single monitor view, all the informations and controls of the SPIDER radio telescope.



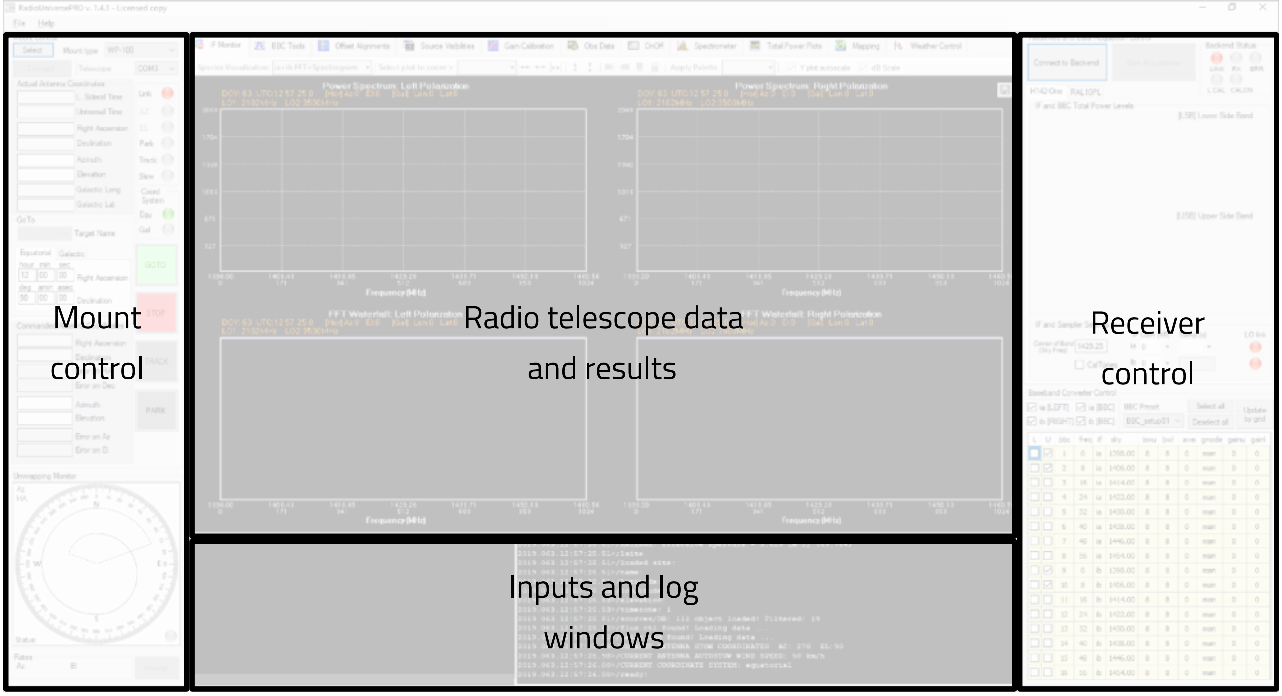
The main window is divided into 4 parts:

- Part 1 (left column): here you find all the data related to the radio telescope mount (both equatorial and altazimuth ones): connection buttons, mount positions, pointed object selection and data, functions buttons, unwrapping monitor (for alt-az mount) and manual movement buttons.

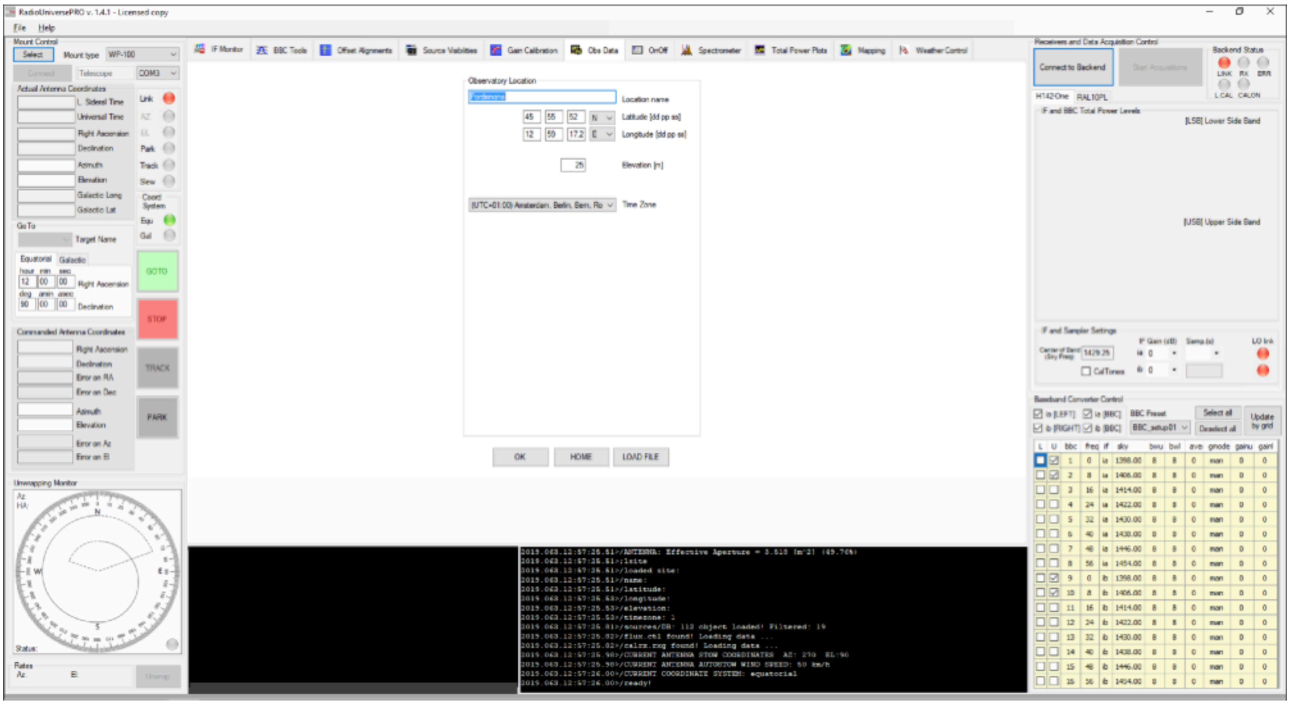
- Part 2 (central area): here you find 10 tabs that give you full access to all the radio telescope features. These will be later described but here you have the tabs list
  - Planetarium (to be added)
  - IF Monitor
  - BBC Tools
  - Offset Alignment
  - Source Visibilities
  - Gain Calibration
  - User Data
  - OnOff
  - Total Power Plots
  - Mapping

- Part 3 (right column): here you find the area related to receivers and data acquisition. Here you can connect to the receiver, have a quick look at BBC Total Power Level (if connected to H142-One receiver), choose receiver settings, activate BBC bands, etc.

- Part 4 (bottom): the bottom part shows the Input Window where you can write manual commands and define your scripts (described later in this manual) and the Log Window there you can see system messages that describe the radio telescope functioning.



Before using the SPIDER radio telescope, you have to insert your location data in the software. Please click on the “Obs Data” tab, this will open this window:



Please insert your Location name, Latitude and Longitude data (in “degrees, minutes, seconds” format), the elevation and time zone. Press the OK button to save the data on the .ini file and use these data for computations.

Connect to the radio telescope mount

Please remember, before connecting to RadioUniversePRO software the mount has to be powered on and connected with the control computer where you have RadioUniversePRO software. Now, on the left column, please click on the Mount Type window and select the mount type your SPIDER radio telescope comes with.

<u>IF YOU HAVE SPIDER 300A OR SPIDER 500A RADIO TELESCOPES</u>

Please power the mount by pressing the power ON button on the RCPU remote control and power unit box). Then in RadioUniversePRO select “WP-100” if you have the SPIDER 300A or “WP-400” if you have the SPIDER 500A radio telescope and press “Connect” button. RadioUniversePRO will connect to the mount and on the left column you now see all the antenna position related data with the controls of the mount itself. If the radio telescope is in stow position, RadioUniversePRO software will show Azimuth=270 and Elevation=90 in the upper part of the left column (Actual Antenna Coordinates area). If you see different values (but you have to be sure that the SPIDER is in stow position, than is pointing the Zenith and with azimuth direction 270 degrees), you need to set the encoders values before moving the radio telescope and this is a very simple task. Just move the mouse in Input Window area and write:

setaz=270 (this command set the azimuth value to 270 degrees, without moving the SPIDER)

setel=90 (this command set the elevation value to 90 degrees, without moving the SPIDER)

Then verify, that, under Actual Antenna Coordinates, you see Azimuth=270 and Elevation=90. Now you can start moving the radio telescope.

The easiest way to check that the radio telescope is correctly moving (to be performed during daytime) is by selecting Sun in the left column and press the GOTO button. The radio telescope will move and point a position close to the Sun. Take a look at the shadow of the feed horn on the antenna. If you can see it quote close to the center (as in the picture), it means that the mount is ok and now it needs to be aligned to the sky.

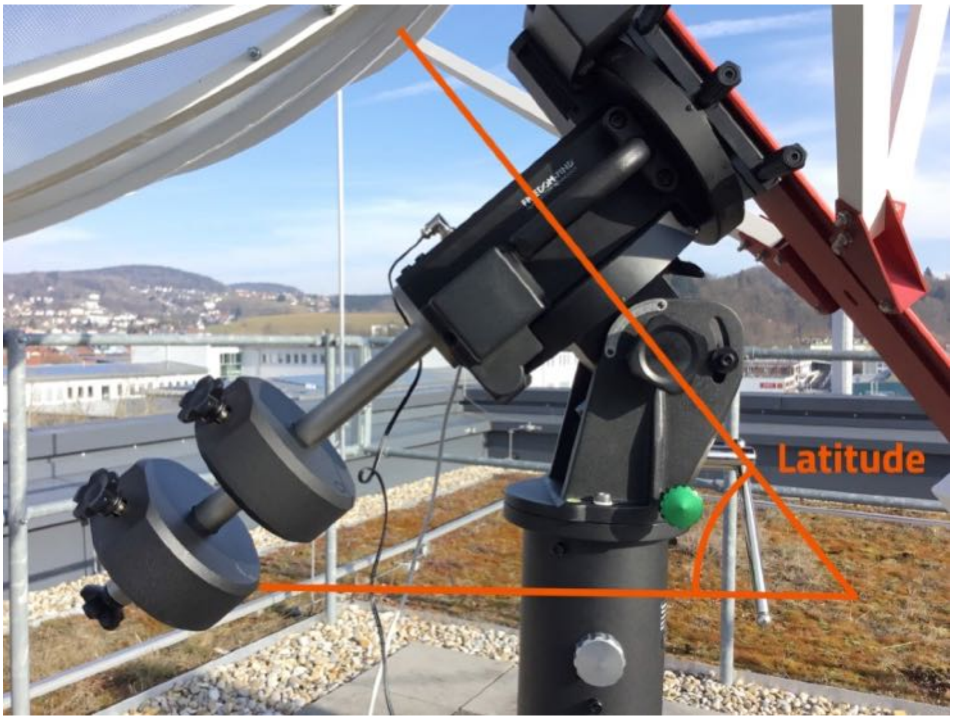
<u>Please go to to the next paragraph to proceed.</u>

IF YOU HAVE THE SPIDER 230C RADIO TELESCOPE

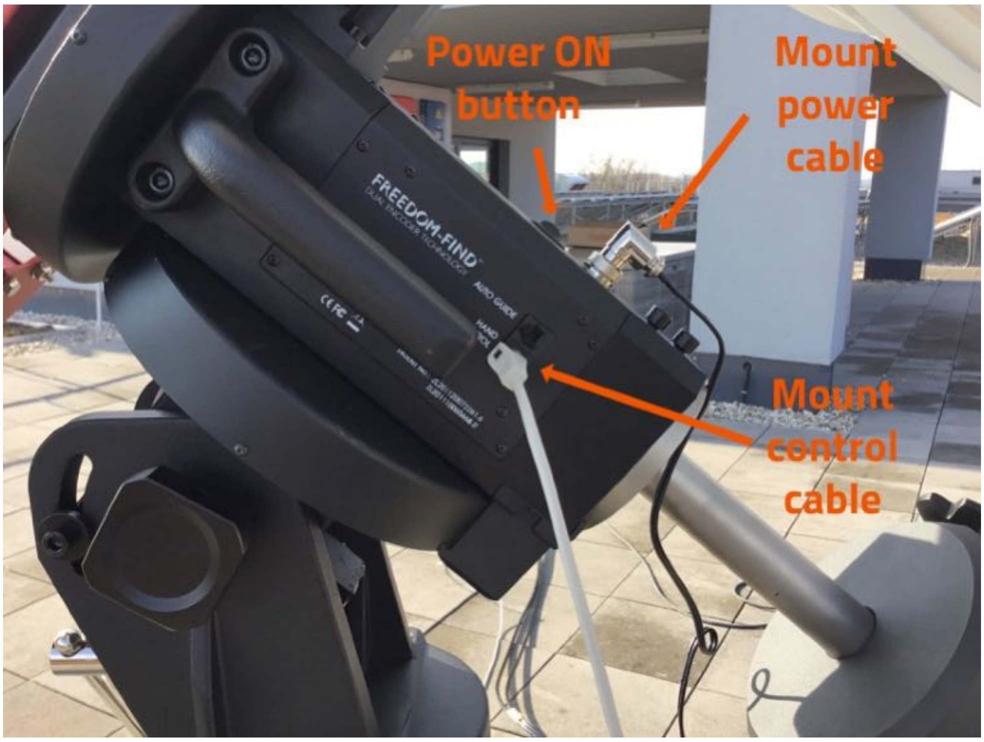
Since the SPIDER 230C radio telescope mount is not weatherproof, you can't leave the radio telescope permanently installed in the field. Please follow this guide in order to have the radio fully operational:

1) Install the antenna on the mount and, having the mount counterweight shaft pointed to the ground (please refer to the mount user manual if you want to know more about counterweights and counterweights shaft), point the antenna to the North direction as precise as possible. If you perform this operation in the night time, you can point the North Star position (if you live in North Emisphere), if you perform the alignment in day time, you can help yourself by using a compass.

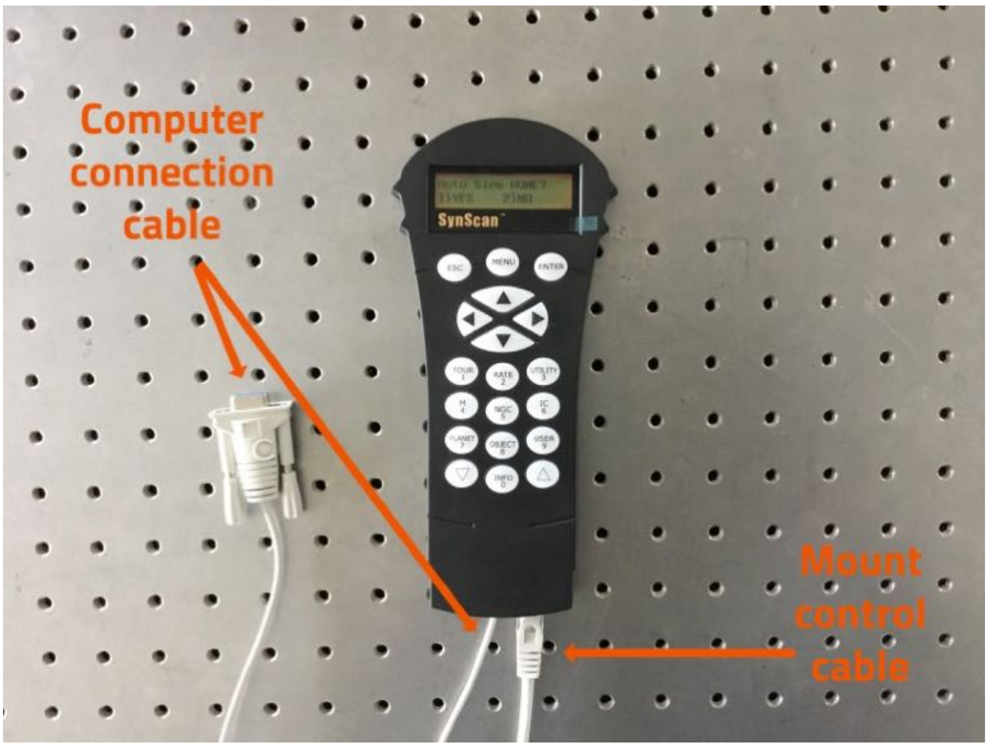
2) Set the mount elevation based on the Latitude of your location. Please use the proper Latitude regulation knob (please refer to the mount user manual if you don't know what is the Latitude regulation knob).



3) Connect the mount to the power cable and then the mount control cable in the proper ports of the mount head, as you can see in the picture below.



4) The mount comes with an handpad that has to be set in the control room, close to the H142-One receiver and the control computer (where you install RadioUniversePRO control software). Please connect the other side connector of the Mount Control Cable to the rightmost port in the bottom part of the handpad. Then connect the Computer connection cable to the other port of the handpad on one side: the same cable has to be connected to a serial RS232 port of your control computer (if you haven't a serial RS232 port, you can use a RS232-to-USB converter, not included standard with the radio telescope).



5) Press the ON button on the mount head.

6) On the handpad you will read the message “Initializing” and then, after a few seconds, “Auto slew home?”, please press button 1 and SPIDER 230C mount will automatically move the antenna until it reaches the HOME position. You see the message “HOME position established”, then press ENTER to confirm.

NOTE: what is the Home Position?
The equatorial mount that comes with the SPIDER 230C radio telescope needs to be aligned to the North Polar Axis (a point in the sky very close to the North Star) in the case the observer is in North Hemisphere. If the observer is in the Southern Hemisphere you need to align for this. When you align the mount to the polar axis, the mount is able to track the objects with a single movement. Polar alignment is a critical factor for an optical telescope but it’s less sensitive in case of a radio telescope because of the less precision requested for a radio telescope tracking and goto. But it’s always important to align SPIDER 230C mount as close as possible to the North (or South) Pole.

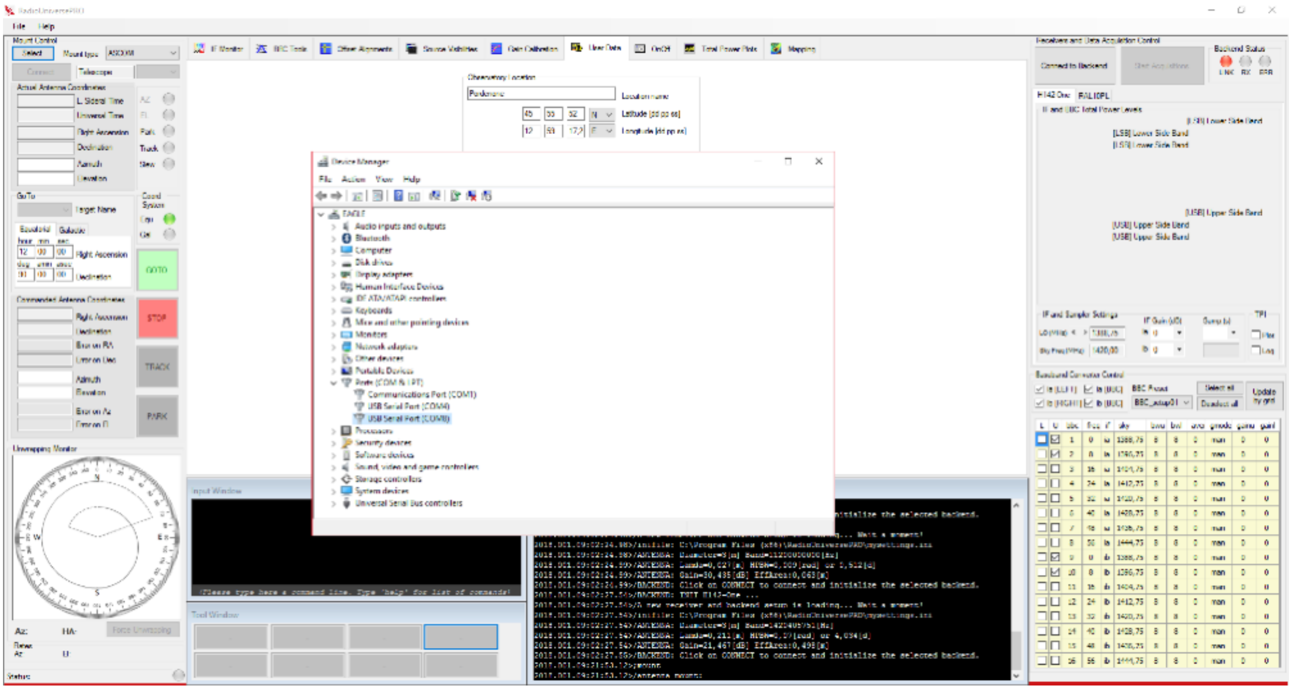
7) A message “Add DEC offset” will appear, press the butt on 2 (NO).

8) The handpad will request you to insert date (USA format, MM/DD/YEAR) and time. Please follow requested data and press ENTER to proceed step by step. At the end you will be asked “Begin alignment?”, please select “NO” and you will see the “Choose Menu”.

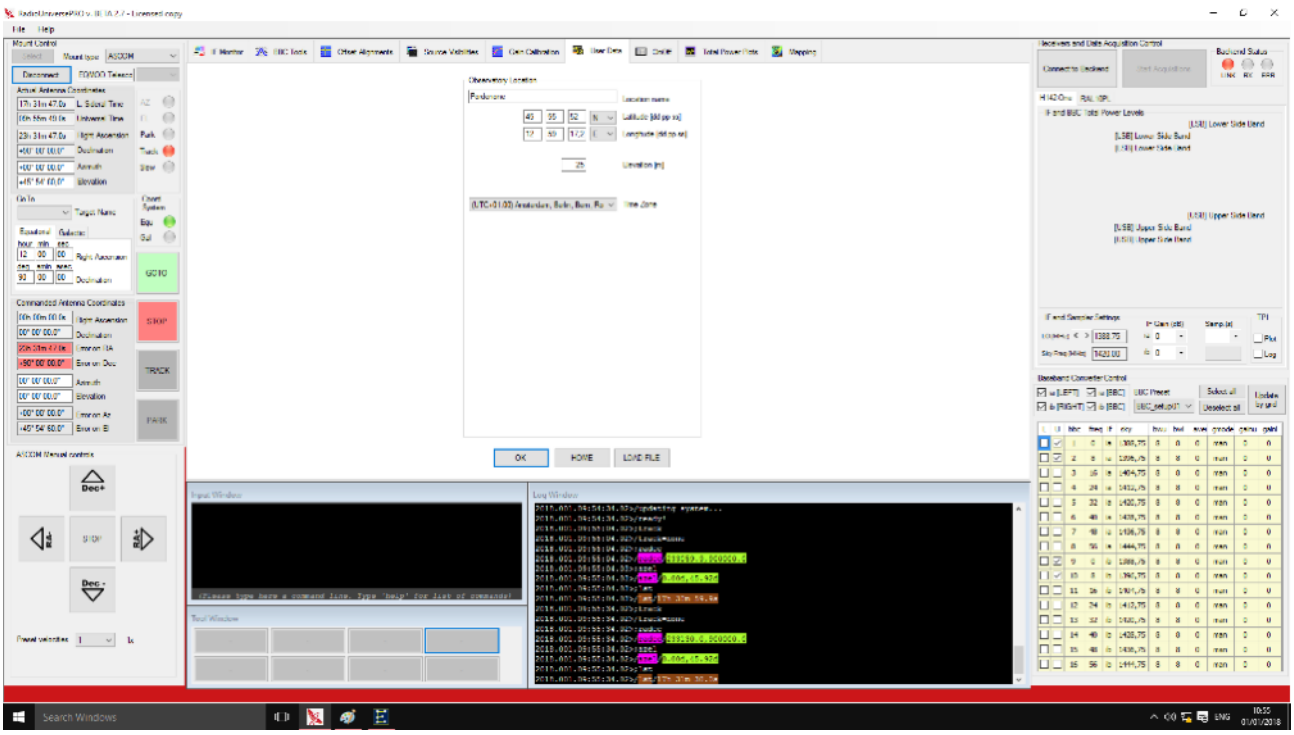
9) Now we have to deselect the use of the internal mount encoders (encoders are not needed for the use of the radio telescope). Press the ARROW UP (the lower right button) in handpad 5 times up to “Auxiliary Encoders” menu, then press ENTER button. With arrow UP or DOWN buttons please select DISABLE, then press ENTER. NOTE: before switching off the radio telescope when you finish using the radio telescope, please remember to activate encoders again or the radio telescope won’t be able to use the AUTO HOME function (and you will have to manually set the radio telescope antenna in HOME position).

10) SPIDER 230C model comes with computerized equatorial mount that can be connected to RadioUniversePRO by using the ASCOM platform (please remember that, in order to control your EQ mount from RadioUniversePRO software, you will need to have already installed in your computer both the “ASCOM Platform” version 6.3 or later and the SkyWatcher ASCOM driver). Select ASCOM in the Mount Type window and then press the Select button. This will open the “ASCOM Telescope Chooser” window, select “SkyWatcher” option, then press Properties button to continue.

11) A new window “ASCOM configuration” will open. Here it’s important to correctly set ASCOM connection parameters to allow RadioUniversePRO to control your EQ mount (please note, now your EQ mount has to be powered ON). In Port option, select the proper COM port the connection cable created when you connected the mount cable to the control computer. If you don’t know what is the port number, please go to Windows “Device Manager” and search for “Ports (COM & LPT)” in the list as you can see in the picture below. By plugging and unplugging the mount cable of to the control computer, you will be able to see the COM port number created by your computer. This is the COM port number you have to select in “Port” option in “ASCOM configuration” window. Then insert the Latitude e Longitude parameters (based on your location) and finally press OK button of “ASCOM configuration” window when done. Finally, press OK button of “ASCOM telescope chooser” window to confirm.

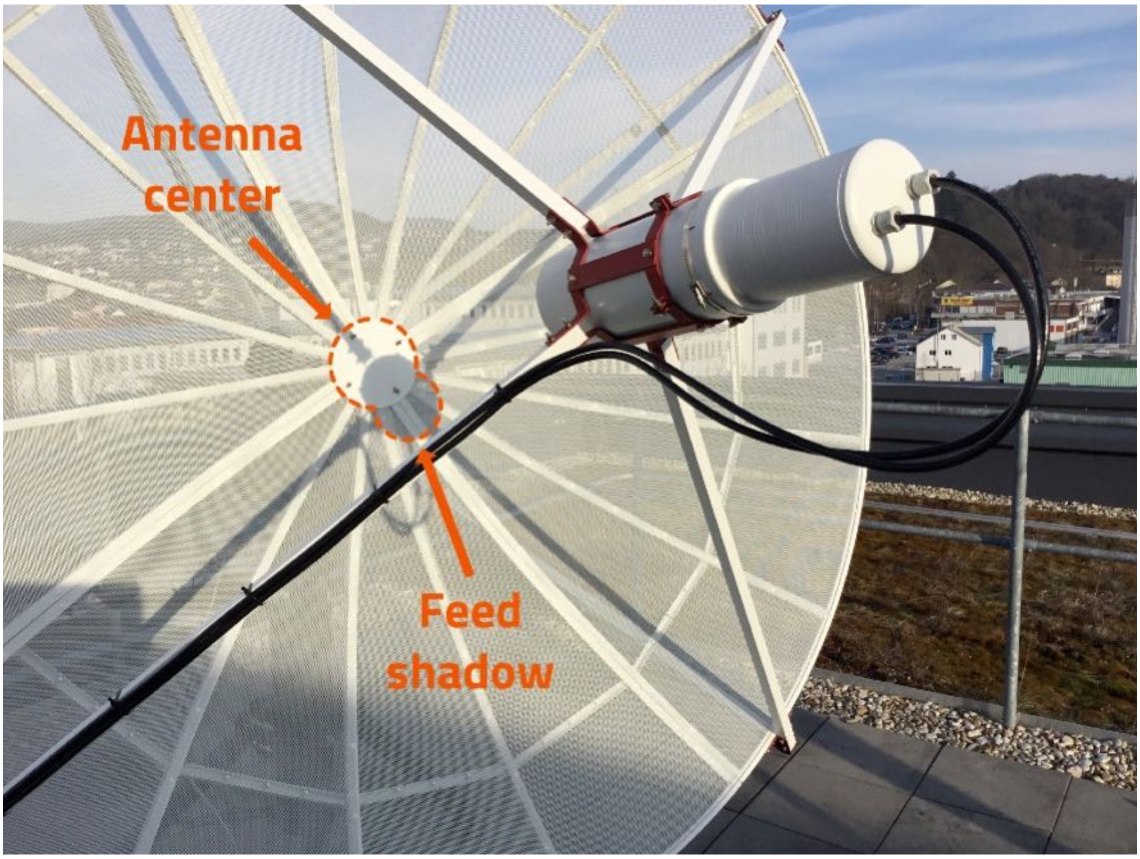


12) Now press the “Connect” button in the up-left part of the mount control area of RadioUniversePRO and the connection to the mount will be stablished (a new window will pop-up, you can hide it because you will have all the mount controls also in the main RadioUniversePRO window). On the left column you now see all the antenna position related data with the controls of the mount itself.

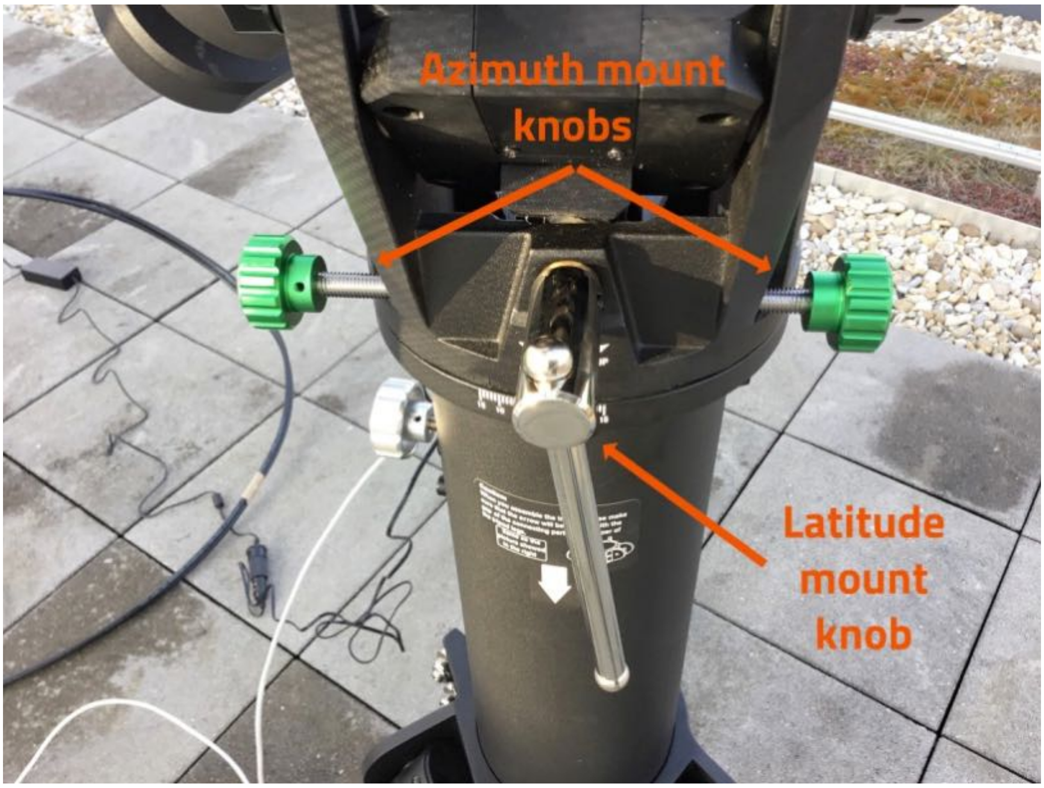


13) Now we have to better align the radio telescope to the sky object position. Please go to SOURCE VISIBILITIES and make a double click on the Sun. The radio telescope will move the antenna close to the Sun. Take a look at the shadow of the feed on the antenna. If it's on the center, it means that the SPIDER 230C mount is already with a good polar alignment. If you see (such as in the picture below) that it's not in the center, please use the elevation and azimuth micro metric controls of the mount to bring the Sun shade in the center of the antenna (as you can see in the pictures shown here).

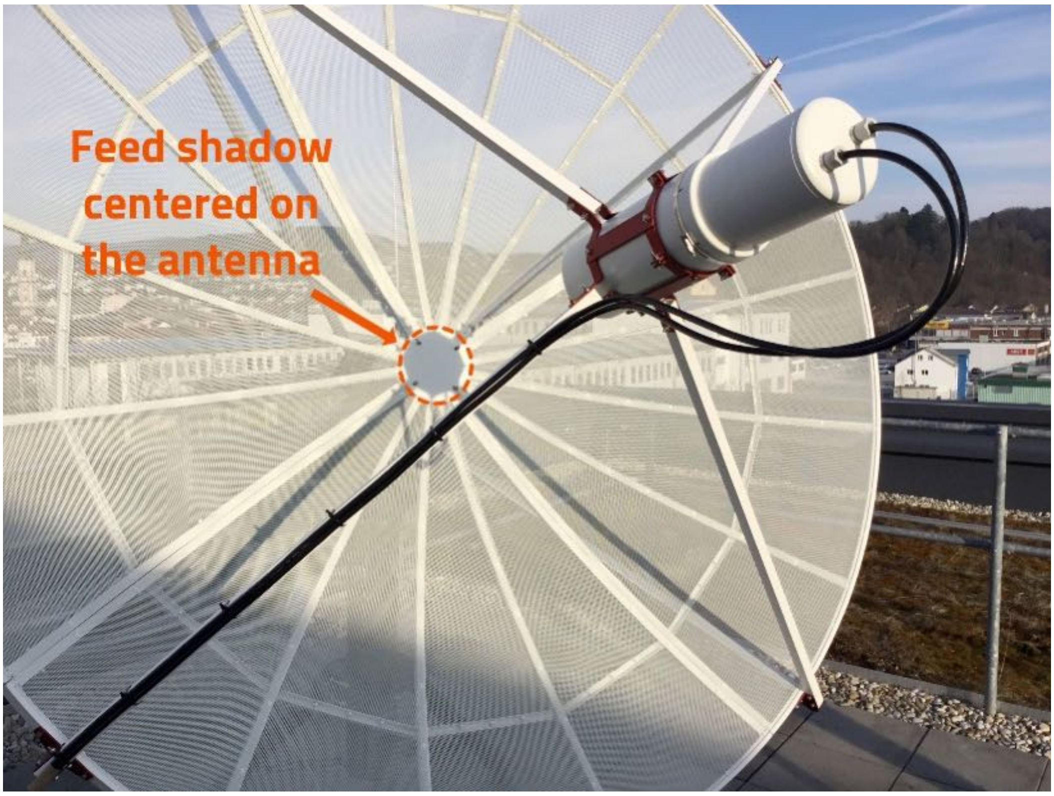
CAUTION:
When you double click on the Sun in Source Visibilities tab, please look at the radio telescope and check it's moving towards the Sun and not in other directions (for example you have to avoid that, for a setting mistake, the antenna points the ground since it may hit something and brake some parts). In this case, just click on the STOP button (in the left column of RadioUniversePRO software) and the radio telescope mount will stop moving the antenna).



14) If you see (such as in the picture on the right) that it's not in the center, please use the elevation and azimuth micro metric controls (shown in the picture below) of the mount to bring the Sun shade in the center of the antenna.

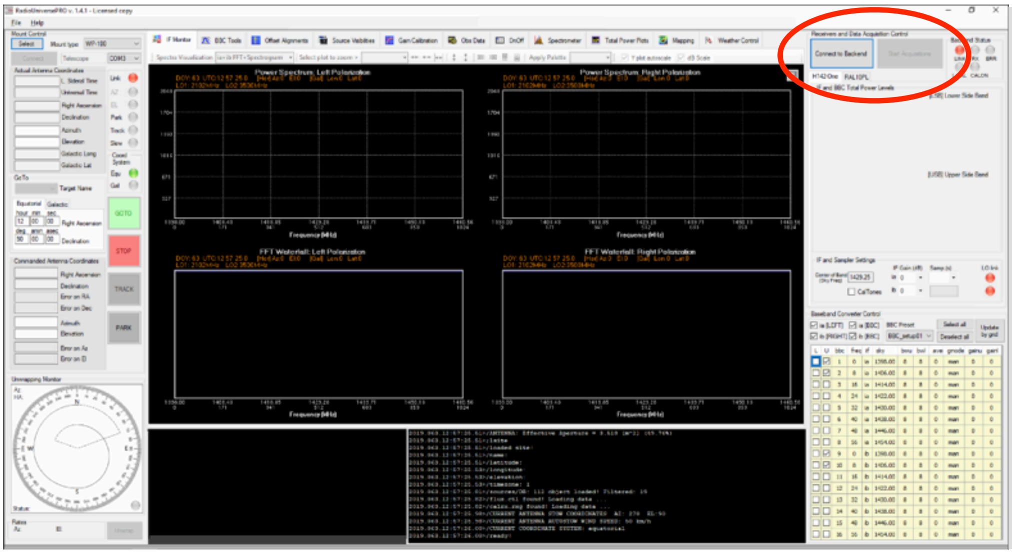


The radio telescope mount is now aligned to the sky and you can proceed with the next paragraph (you will be able to refine the radio telescope alignment to the radio sources in the sky by using the “Offset Alignment” tool of RadioUniversePRO as described in the next paragraphs).



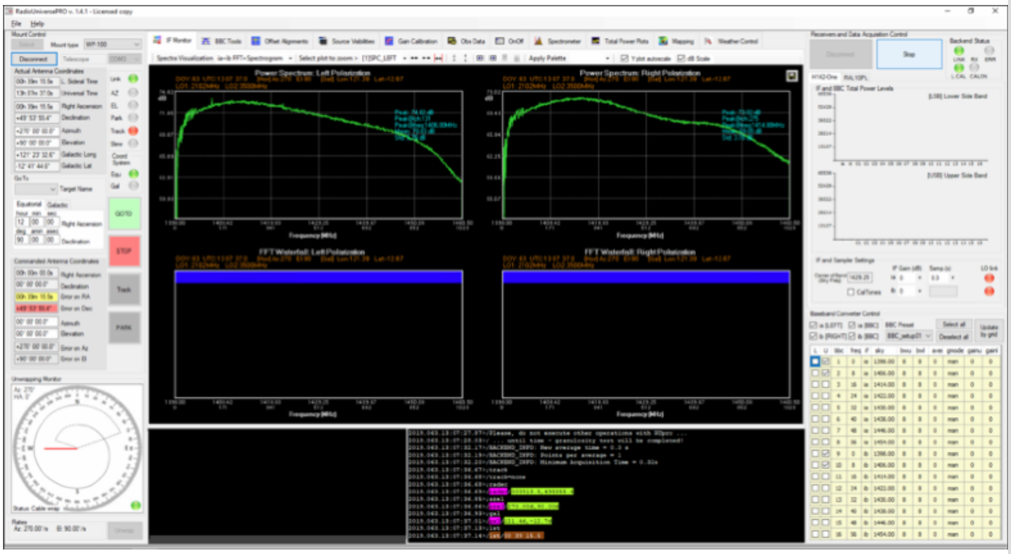
Connect to the radio telescope receiver

Before connecting to RadioUniversePRO software the receiver has to be powered on and connected with the control computer where you have RadioUniversePRO software. In the Receiver control area (on the right column) please press the “Connect to backend” button and, in the Log Window area you will see connection status messages to the receiver. Please remember that, in order to connect to the H142-One receiver, your control computer has to have a static IP number in the 192.168.1.x range (receiver is 192.168.1.101).



NOTE: based on your country, you may have a power converter to properly power the receiver. For example, for USA there is a unit that converts 110V to the 220V requested by the receiver to work. Please turn on this converter FIRST, then you can power the receiver box. When you finish with the SPIDER, please turn off the receiver box and then the converter.

Then select “Start acquisitions” button: the IF monitor will display the data coming from the radio telescope to the receiver in form of a Power Spectrum (upper row) and FFT Waterfall (lower row), both for Left and Right Polarization. Power spectrum is the graphical representation of the electromagnetic power distribution in the operative band, through the RF chain. The spectrum graph shows, for every channel, signal spectral power: that is, the levels are proportional to the square of the input signal of the FFT execution. The power spectrum is then used to calculate the total power use through the integral of the entire bandwidth on the left and right channels. Total power values are reported on the right column graph “IF and BBC Total Power level” with the labels “ia” and “ib” that indicate the values “Intermediate Channel A” and “Intermediate Channel B” (IF left and IF right).

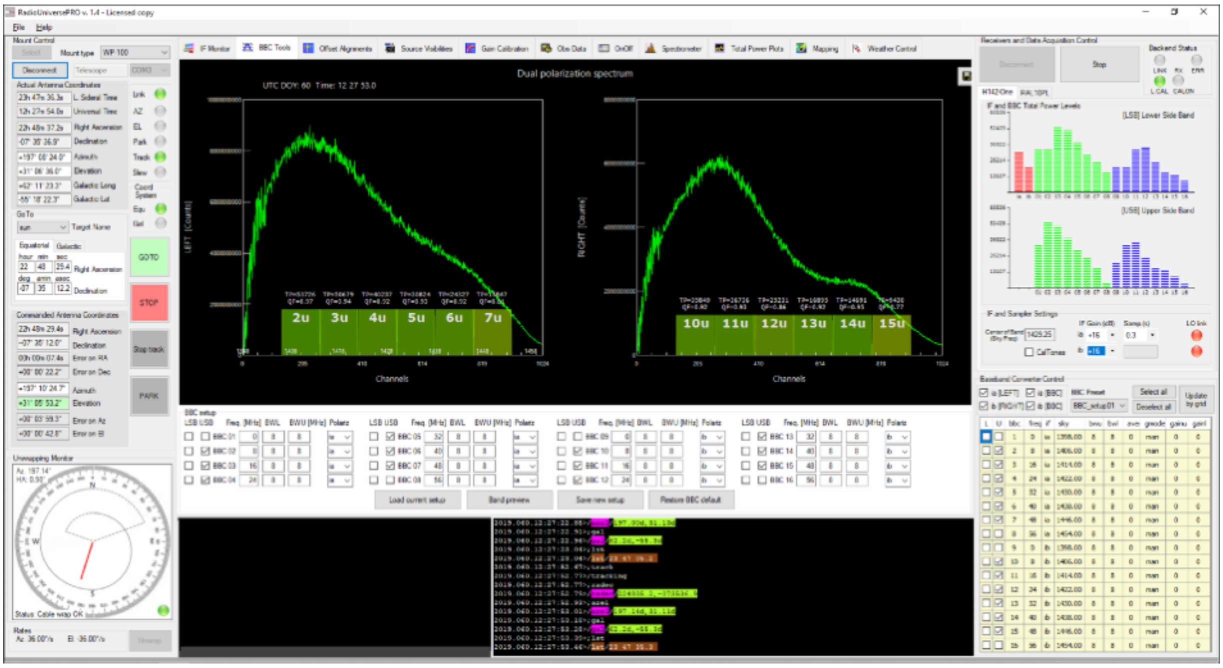


In the upper part of the window you have the controls that allow you to better visualize the signal. Please deselect “Y plot autoscale” option and then use the buttons

to increase or decrease the zoom factor, the scale position (both X and Y axis) for the left (SPC_LEFT) or for the right (SPC_RIGHT) polarization.

BBC Tool

BBC Tools window allow you to visualize in real time the uncalibrated power spectrum of the input signal in both IF left and right. Often we can find interferences in the IF or, more simply, there may be parts of the spectrum that we don't want to consider during the measurements. For this reason RadioUniversePRO allows you to use a group of digital filters (16+16) fully tunable on the 2 Intermediate Frequencies. Every filter is identified by a BBC (Base Band Converter) label and a number from 01 to 16. Every filter can be set in frequency (starting frequency in base band in the IF), in bandpass (the width of the filter band and that will define it's radiometric sensitivity and the "exposition" to RFIs) and you can choose the IF to whom to associate it. For example the BBC02 can be associated with the IF left or right by using the options in the lower part of the BBC Tools window. RadioUniversePRO starts with a default configuration that can be fully modified and saved (by clicking the "Saved new setup" button) for the future use.



Every RadioUniversePRO procedure (like creating a radiomap, create a total power plot, perform offset alignments on a radio source or the gain calibration) can be performed on a precise BBC setup in order to let you optimize the used bandwidth (by excluding RFIs in band or centering one or more filters on a particular region of the spectrum for the Hydrogen study).

BBC filters are very important to remove unwanted signals from recordings. In order to do so, look at the real time spectrum and search for very narrow signals, if any. These will be most probably artificial interferences (but please not to confuse with the 1420 MHz neutral Hydrogen line!). In order to personalize the BBC setup, please follow this procedure:

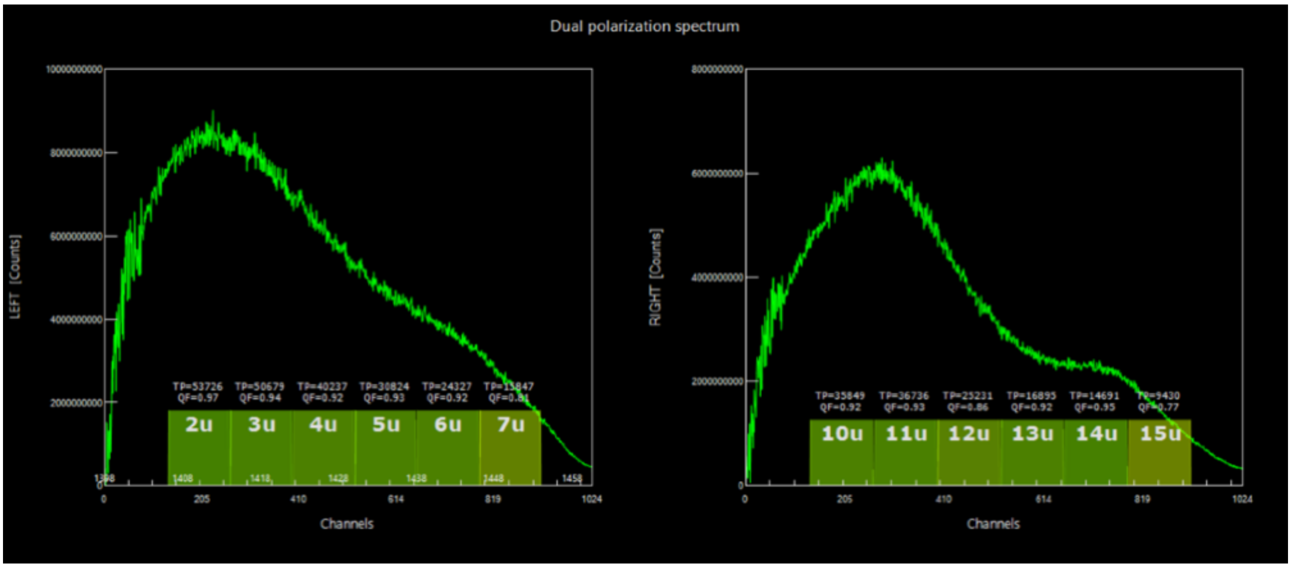
1) Activate or deactivate one or more BBC filters (as you can see in the picture above) by selecting the corresponding USB option
2) Every BBC filter can be personalized: you can change "Freq.", "BWL" and "BWU" values for every filter and then press ENTER in the keyboard to apply. You will see the position of the corresponding filter changing just below the spectrum.
3) When you find the BBC setup that best fits your system, click on "Save new setup" button to apply.

NOTE:
Since SPIDER radio telescope antenna has a high directivity, we suggest you to check BBC filters and the absence of artificial signal every time you move the radio telescope to a new object, before capture.

BBC Tool is a very important feature that helps you also to correctly set the gain value for the left (ia) and right (ib) polarizations. In fact, when you record data from the Sun, the radio emission is so strong that you don't need to change the digital gain. But when you want to record data from all the other objects (like Cassiopea A, Cygnus A or Taurus A) you have to increase ia and ib digital gain in order to better match with the system dynamic. In order to do so, we can use the BBC Tool feature and follow this procedure:

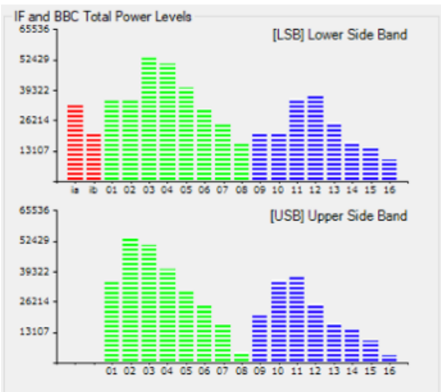
1) point the radio telescope to the object you want to record.

2) take a look at the TP levels as reported for every BBC filter.



3) increase the gain for ia (left) and ib (right) polarization in order to increase the value of TP to around 20000 counts. Please note, avoid too high values because at 65536 counts the BBC filter will be saturated and it will be impossible to record data.

4) You can always have a quick view of the TPI levels by looking to the graph on the right of RadioUniversePRO software, showing both LSB and USB. In red you have the ia and ib TPI levels. In green you have the left polarization, blue the right one.



NOTE:
Since the IF isn't a flat curve, TPI values for BBC filters may change a lot given a flat digital gain. Please choose a digital gain that offers good and not saturated TPI values for all the BBC filters.

In order to find the best value, after you have the radio telescope tracking the sources move the radio telescope in off position with the command: azeloff=7,7

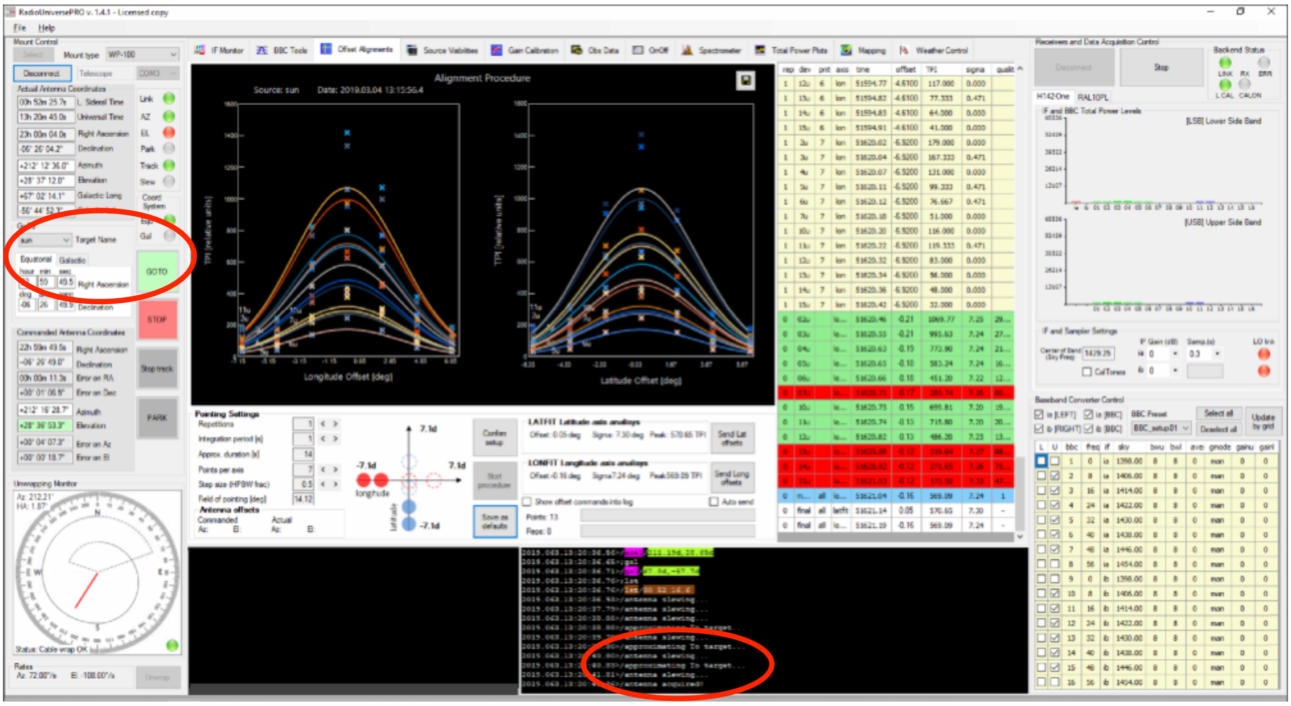
Check for the TPI values and some back to the source (on) position with the command: azeloff=0,0

By comparing TPI counts in ON and OFF positions, you can define the best gain value for your system.

5) Now that you correctly set the digital gain, you can align the radio telescope and capture data.

Offset Alignments

Offset alignments is the instrument that is used to synchronize the radio telescope on the radio sources position on the sky, by reducing the pointing errors on 2 axis of the mount. Before using this tool, you need to point a radio source (by selecting on the “GoTo” window on the left column and clicking on GOTO button or by double-click on one of the listed radio sources in the “Source visibilities” tab). Usually you can use the Sun as a reference radio source since the signal is very strong. On the left “Mount control” column please select Sun as “Target name” and press GOTO button. Wait for the radio telescope to point the Sun, you will see the “Antenna acquired” command in the Log window.



Then, after you selected the proper BBC you want to use, you can design the alignment settings:

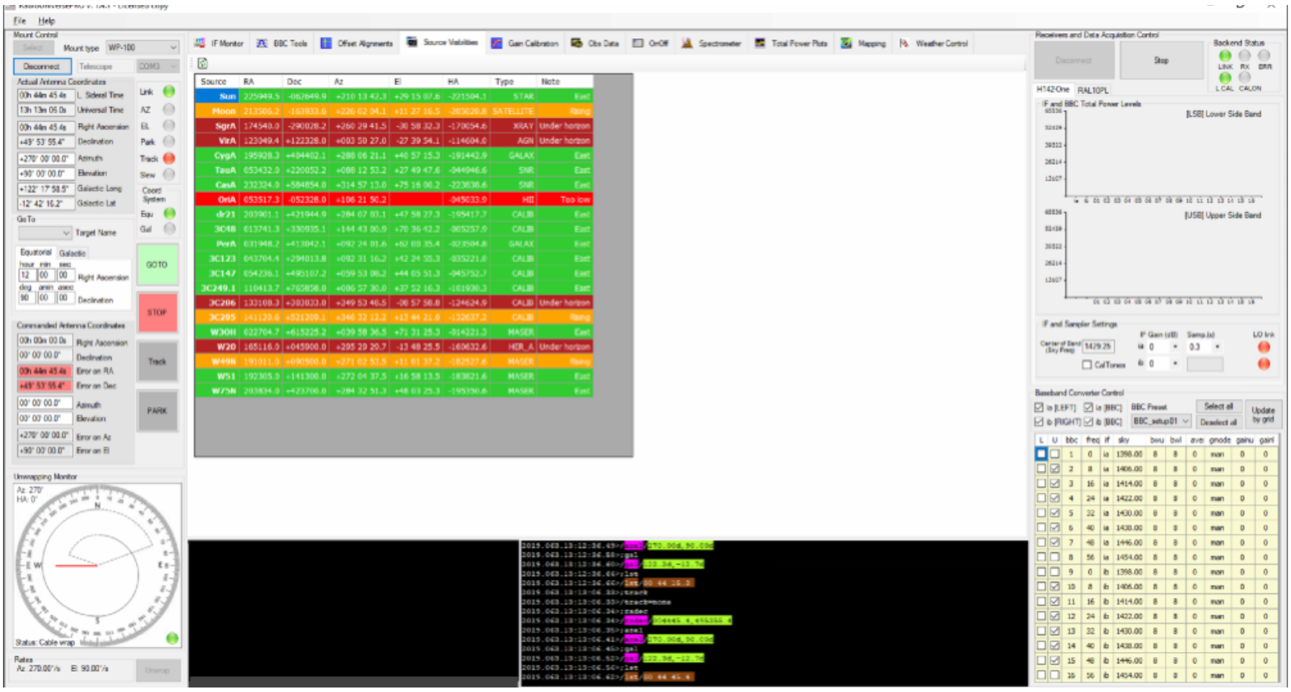
1) Repetitions: how many times the alignment procedure has to be executed (default: 1)
2) Integration period (s): how many seconds the radio telescope has to measure for every point. If you align on the Sun, 1 second is ok. If you align on weaker objects, you can use 5 or 10 seconds for a better S/N.
3) Points per axis: how many points you will have per axis. If you’re not very close to the object, you will set a higher number and the radio telescope will move longer to search for the radio source.
4) Step size (HPWD frac): step size between recorded points. Usually 0.5 is good to start, 0.2-0.3 will be used if you want to make another alignment to refine it.

To start, press the “Confirm setup” button and then the “Start procedure” button. The radio telescope will perform automatic measurements of Total Power values by varying latitude and longitude offsets (please make sure the object you choose has a good detectability with the SPIDER radio telescope you’re using since this will allow a better calculation of the offset alignment). When the procedure is completed, RadioUniversePRO will complete plotting of the graphs and calculate the best offsets to send to the mount electronics in order to better align to the pointed radio source. When the calculation is performed, pleased press “Send Lat offset” and “Send Long offset” buttons to confirm the values. If you want to save this values as defaults, pleas press the “Save as default” buttons.

TIP: after you complete the first alignment, you can restart a new one to refine it. You can reduce the points number and the step size in order to be more accurate in the offset calculations.

Source Visibilities

Source Visibilities tab lists the most powerful radio sources in the sky. This tab is designed in order to allow you to have a quick idea on the radio sources available that you can point and study with your radio telescope (detection level is different based on the SPIDER radio telescope model you use). In order to point the radio telescope to any of the listed radio sources, please make a mouse double click on the radio source row.



For every radio source are listed
- RA (Right Ascension) coordinate
- DEC (Declination) coordinate
- Az (Azimuth) value
- El (Elevation) value
- Type: star, satellite, AGN (Active Galactic Nuclei), XRAY (X ray source), SNR (Super Nova Remnant), Galaxy, HII (Hydrogen II source), CALIB (Calibration source)
- Note: direction, if it's too low to study, it it's under horizon, transiting or setting.

Every radio source is highlighted with a color:
- Green: visible or transiting (to let you immediately see the objects close to the Zenith)
- Orange: setting
- Red: too low for a detection
- Dark red: under horizon

Gain Calibration

NOTE: Gain Calibration procedure requires the use of the optional Noise Generator. If you start the Gain Calibration procedure without the optional Noise Generator installed in your SPIDER radio telescope, the shown results won’t be valid.

Gain Calibration procedure executes Total Power measurements (by using the proper BBC filters the users select in advance) on a set of radio sources with a known radio flux by literature. These radio sources are listed in the “flux.ctl” file in the RadioUniversePRO folder and this file may be edited and updated based on user requirements. During the Gain Calibration procedure, a proper polinome function allow RadioUniversePRO to calculate the theoretical flux in Jy the radio telescope would record without any atmospheric attenuation or any gain loss because of different factors (antenna deformation because of its weight for example).

Having selected the radio sources to calibrate to, Gain Calibration procedure automatically schedules pointing on them all, on different elevations, always considering the theoretical flux in Jy. The, for every radio sources these Total Power measurements will be performed:

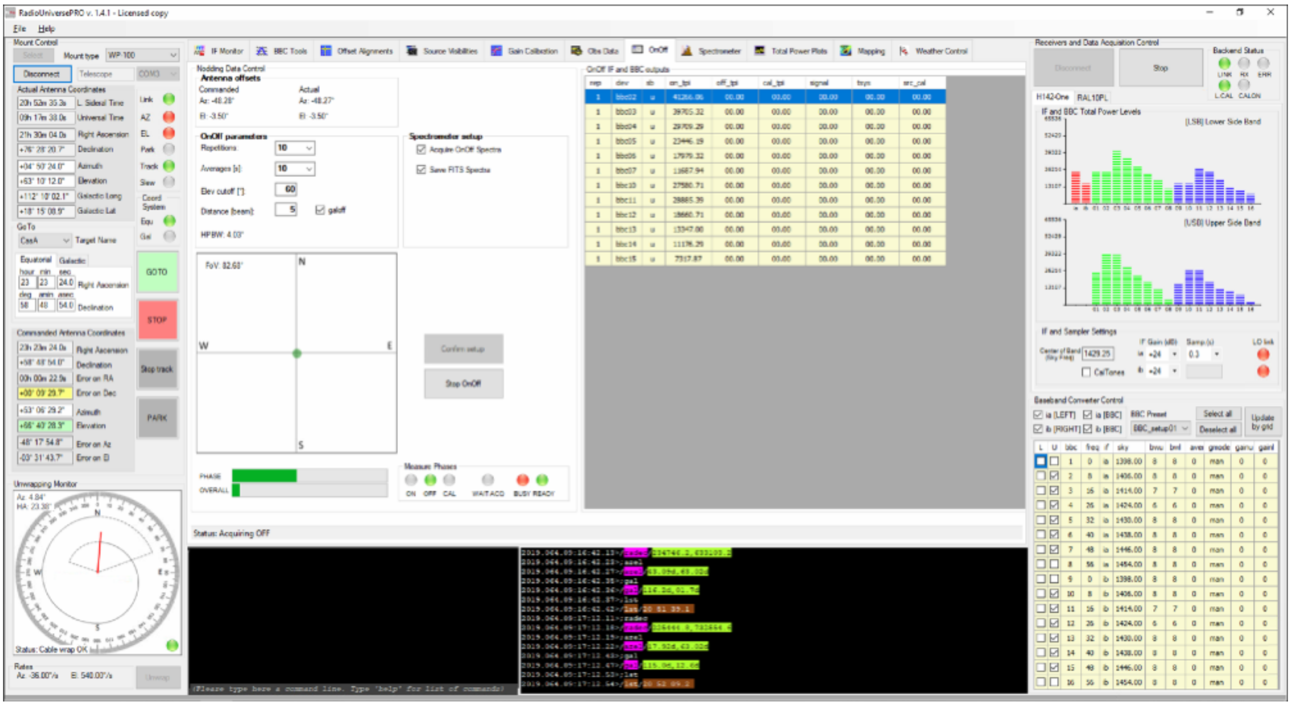
- ON source (on the radio source, with noise generator OFF)
- OFF source (off radio source with noise generator OFF)
- CAL (off radio source with noise generator ON)

Measurements recorded in every step are used to calibrate the noise generator and the radio telescope and this will allow to record antenna temperature and flux measurements.

OnOff

OnOff tab allow you to perform an ON-OFF recording on a radio source. The radio telescope is pointed to the radio source (ON position) and the data is collected. Then it's moved in a position OFF the source and another set of data is collected and used to calibrate the previous one. This way you're able to reduce the radio noise and the effect of external components (like the Earth atmosphere). The ON position is defined by the object selected in the "GoTo" window of the left column of RadioUniversePRO. In the "OnOff parameters" area you can select:

- Repetitions: the number of On-Off measurements the radio telescope has to perform.
- Averages: time in seconds the radio telescope has to stay ON source and OFF course (to record and average data)
- Elev cutoff (°): if the radio telescope is pointed at a radio source with elevation higher than this value, it moves in elevation to reach off position. If it has a lower value, it moves in azimuth.
- Distance (beam): the distance in beams number from ON to OFF position
- galoff: the OFF point position will be calculated in Galactic coordinates. Please select this option if you want to record neutral Hydrogen line in Milky Way plane since the OFF position will be set in "n" distance beams from the galactic plane.

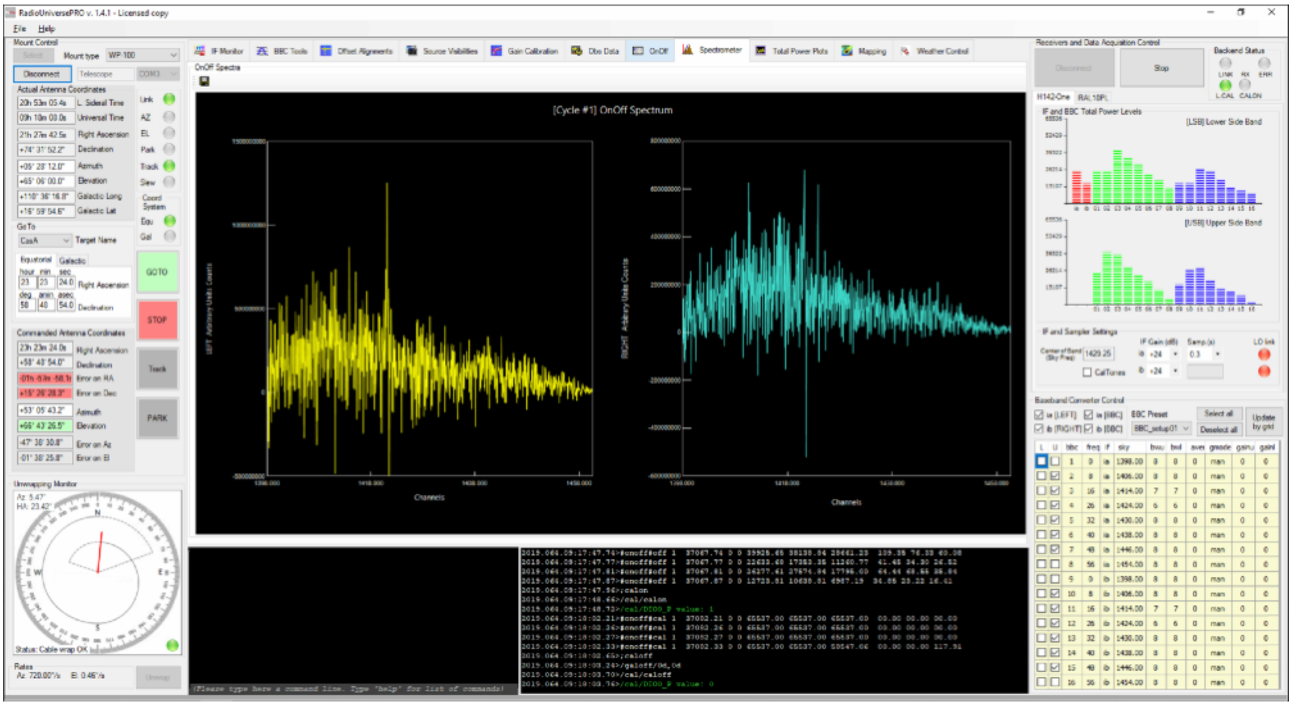


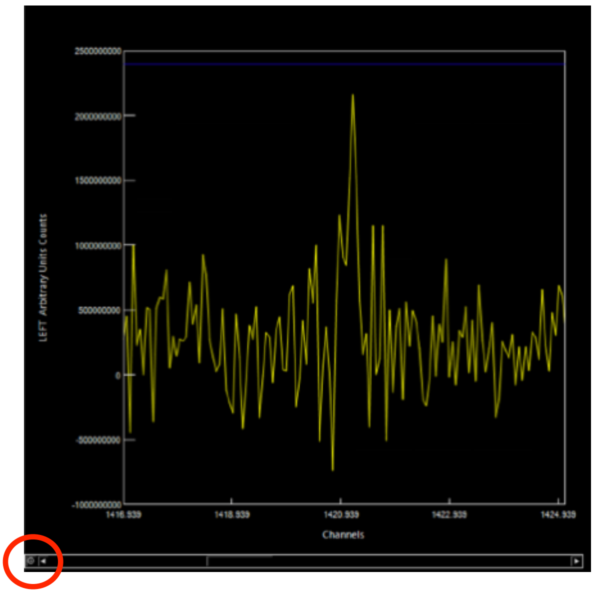
If you want to record also spectra, just select "Acquire OnOff Spectra" option. If you want to save data in FITS format (it will be saved in c:/programs/RadioUniversePRO folder), please select the "Save FITS Spectra" option. Before starting recording data, please click on "Confirm Setup" button and RadioUniversePRO will verify if the input values are corrected. Then click on "Start OnOff" button and the radio telescope will perform the requested measurement. In the central part of the window you can find the "Antenna Offsets - Commanded and Actual". This values indicate how much the radio telescope is off axis during recordings.

TIP: if you want to record data from weak objects (different from the Sun) we suggest to select 10 repetitions and at least 10 seconds in the Average option in order to increase the signal to noise ratio.

During the ON-OFF sequence, RadioUniversePRO software shows you:

- Phase: duration of each step in ON-OFF procedure
- Overall: duration of the entire ON-OFF procedure
- Measure phases: this indicates if the radio telescope is recording the ON, OFF, or Calibration points. RadioUniversePRO also displays on the graph the position of the OFF point (red) in respect of the ON one (green).





During acquisition, every spectrum is shown in the "Spectrometer" tab, both for left and right polarizations. At the end of acquisition, RadioUniversePRO will automatically perform an average of the data and will display the spectrum.

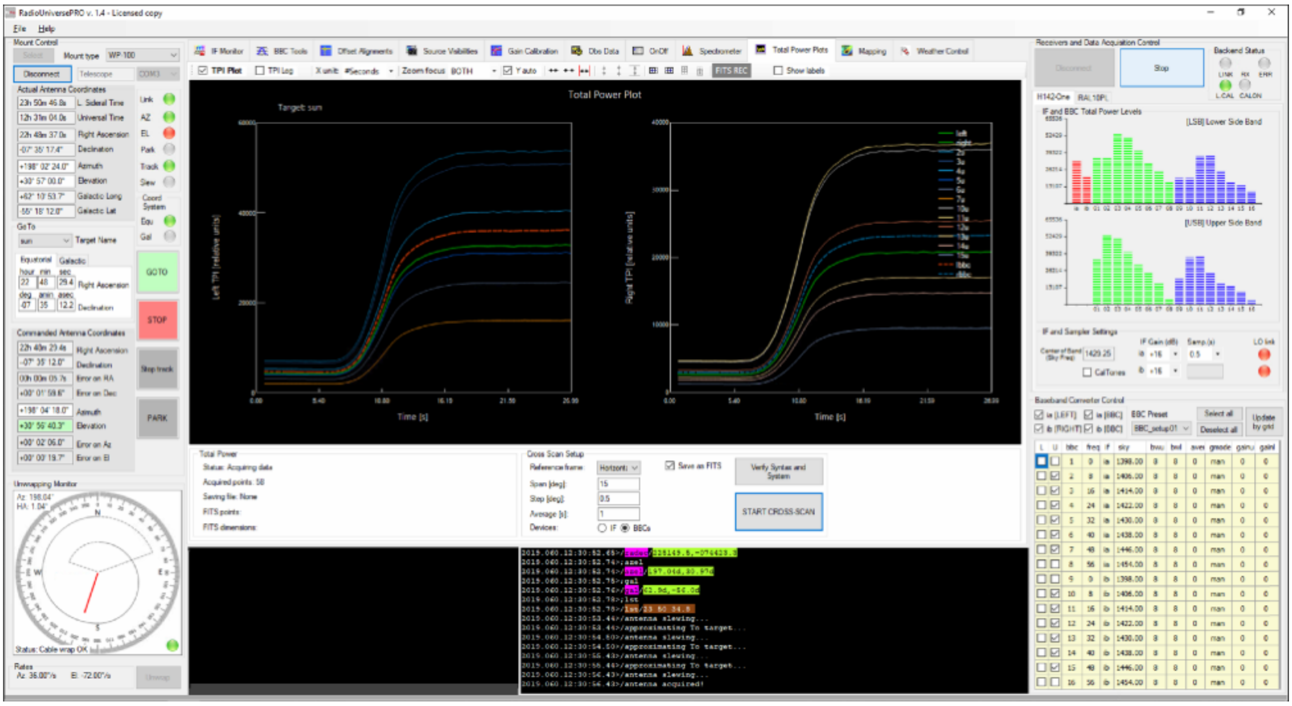
During data acquisition, you can zoom in the spectrum to better visualize parts of it. In order to do so, move the mouse into the spectrum window (you can choose the left or right polarization) you want to start the zoom in (for example, if you want to better see the neutral Hydrogen line at 1420 MHz put the mouse around 1415 MHz). Then click the left button of the mouse and, keeping the mouse clicked, drag the band up to the higher level you want to visualize (for example 1425 MHz). Release the mouse and the graph will be zoomed in. In order to reset the view, please click on the round button to the left.

At the end of the recordings, your data will be saved in c:/programs/RadioUniversePRO folder, also in FITS format if you selected the relative option.

TIP: if you want to record neutral Hydrogen line at 1420 MHz, we suggest so set the distance beam at least at 4 in order to avoid to have, in the OFF point, still some signal from the neutral Hydrogen line.

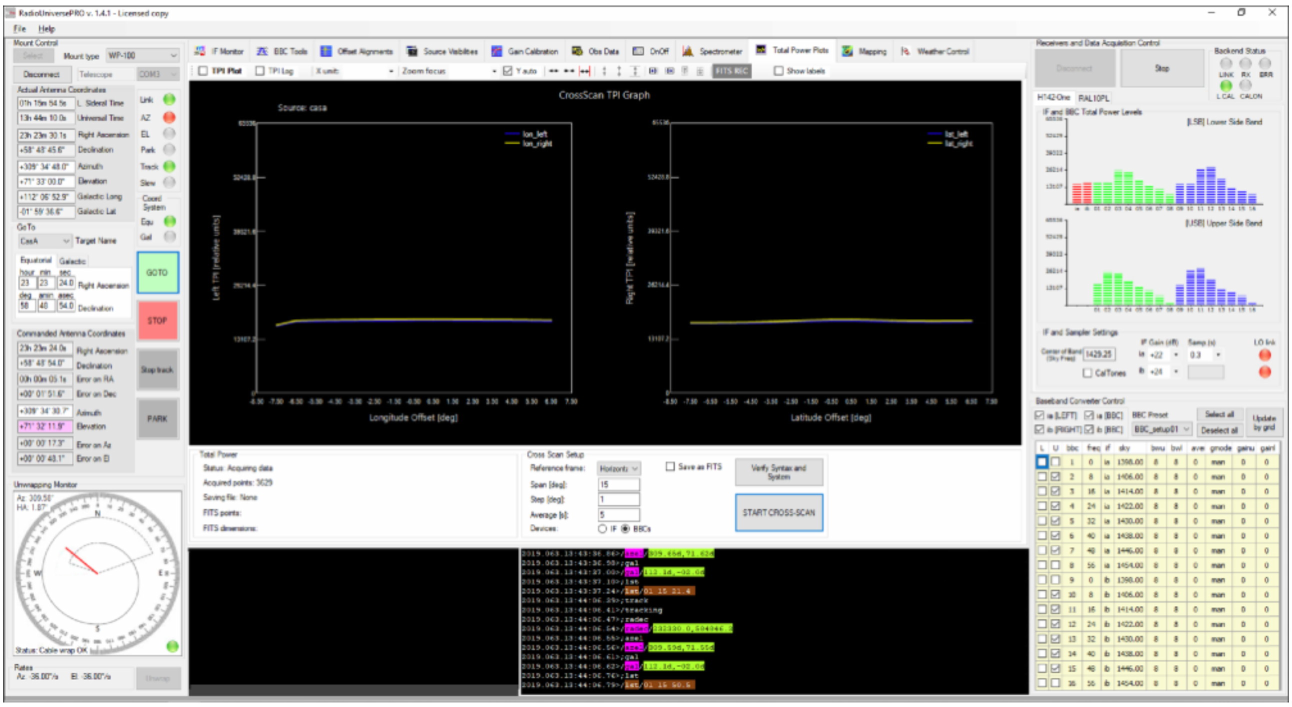
Total Power Plots and Cross Scans

Total Power Plots are the perfect instruments to easily visualize and record the variation of the radio signal flux over time. Just select the “Plot” option and you will see the value in counts for every BBC filter plotted over time. By selecting on "Show Labels”, RadioUniversePRO will add the BBC filters names in the graph.



In the Total Power Plots tab you have the controls that allow the radio telescope to automatically perform a Cross Scan. In the lower right part of the window you can find the settings:

1) Reference frame: horizontal, equatorial or galactic.
2) Span (deg): the length in degrees of the Cross Scan you want to record.
3) Step (deg): the step in degrees between points in the Cross Scan.
4) Average (s): the number of seconds the radio telescope has to record data for every point.
5) Devices: IF (to save the entire bandwidth) or BBC (to save only with the selected BBC filters).

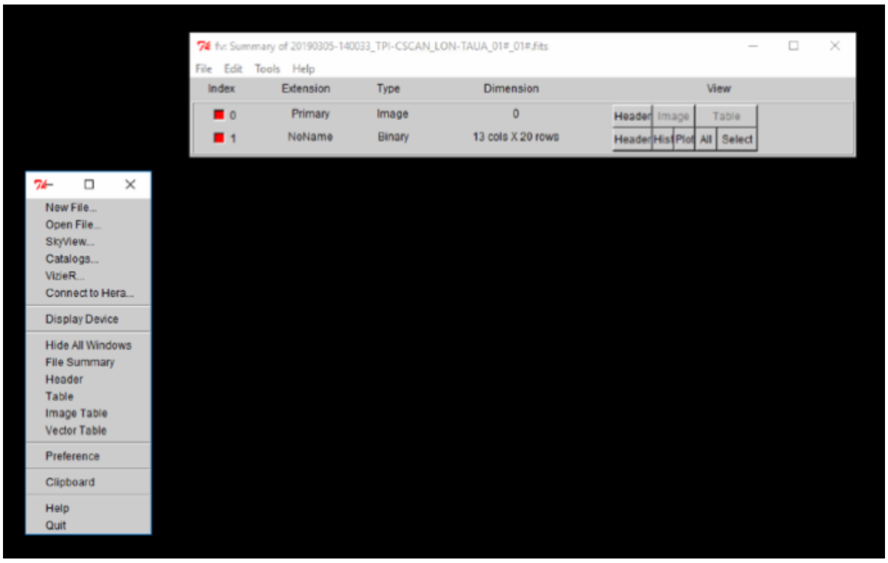


TIP: a Cross Scan recorded on the Sun is a very good and easy way to verify pointing accuracy, focus position and the presence of secondary lobes. Suggested parameters for the Cross Scan on the Sun:

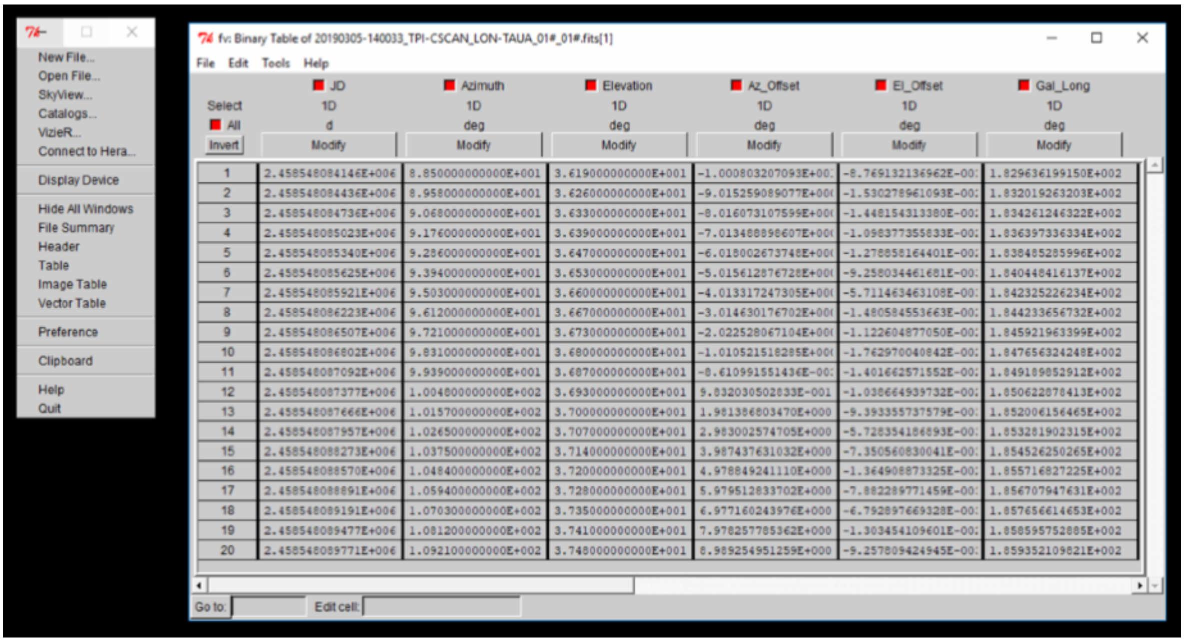
1) Reference frame: horizontal
2) Span (deg): 15
3) Step (deg): 0.3
4) Average (s): 1
5) Devices: BBC

If you want to record data from weaker objects (like Cassiopea A or Cygnus A) you have to increase a lot the Average value, at least 10 seconds, in order to increase signal to noise ratio for every point.

If you select the "Save as FITS" option, RadioUniversePRO will save the Cross Scan in FITS format for a later processing. In order to start Cross Scan acquisition, please press "Verify Syntax and System" button and then the "START CROSS SCAN" button. You will see the data updating in the graph, plotted for the 2 axis.



If you select the "Save as FITS" option, you can process data by using the fv FITS Viewer (available for Linux, macOS and Windows) you can download from the page https://heasarc.nasa.gov/ftools/fv/. After you install, please open it, you will see this interface. In order to open your FITS file, please select "New File" and open the FITS file in the c:/programs/RadioUniversePRO folder. You will see the new Summary window. If you click the ALL button, you will see a new window with all the recorded data. Here you can find these columns:

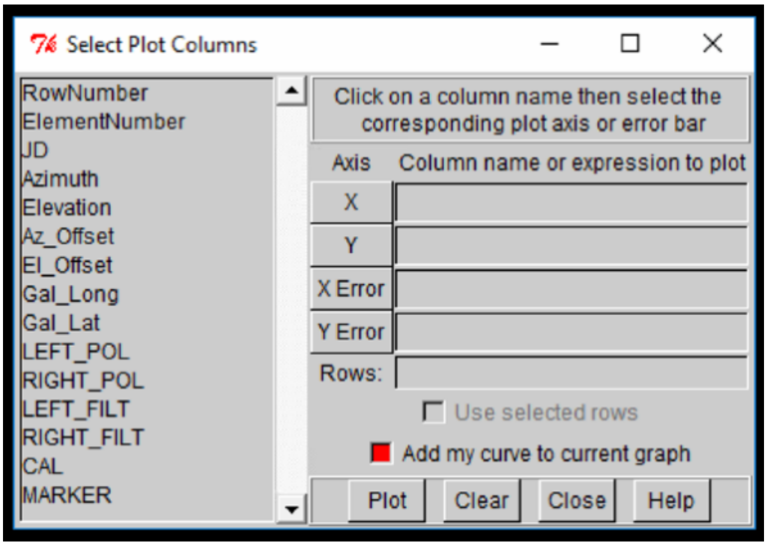


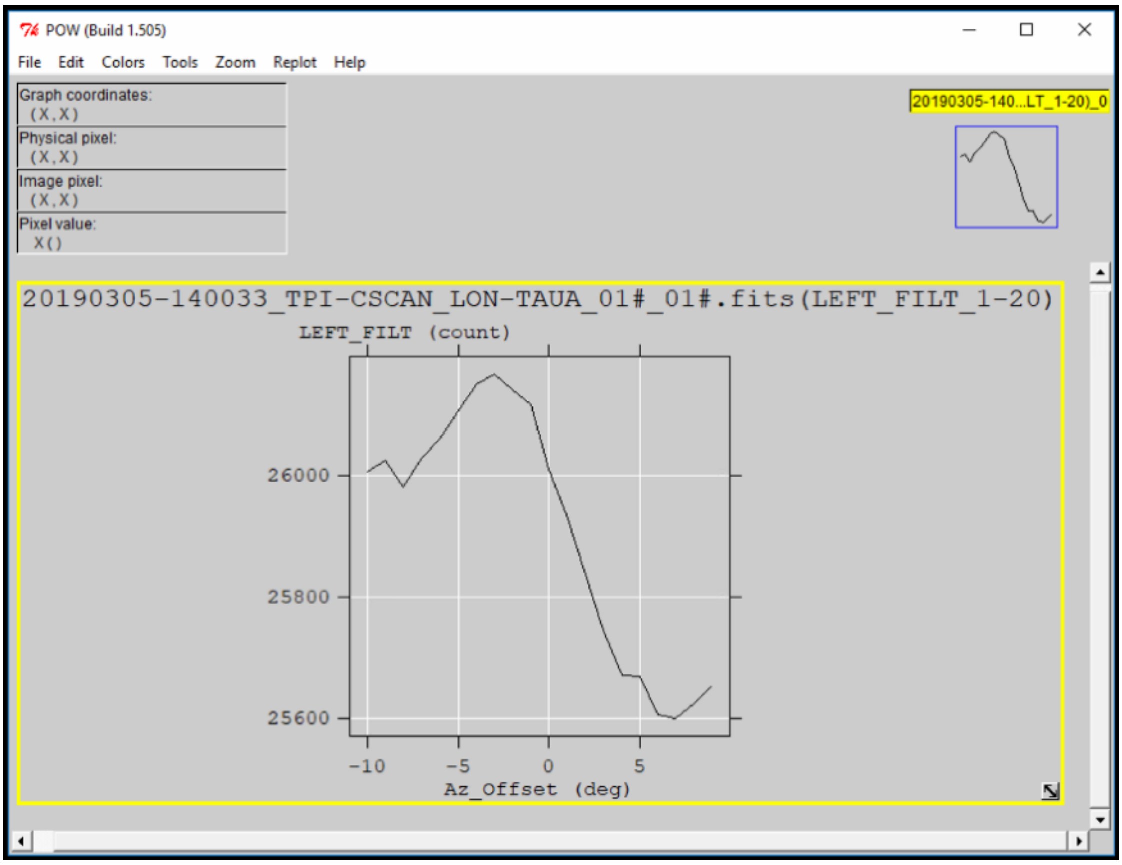
1) JD, time in Julian Date
2) Azimuth
3) Elevation
4) AZ_Offset: offset in Azimuth in respect to the pointed object
5) El_Offset: offset in Elevation in respect to the pointed object
6) Gal_Long: longitude in galactic coordinates
7) Gal_Lat: latitude in galactic coordinates
8) LEFT_pol: total power in counts in the left polarization and entire IF
9) RIGHT_pol: total power in counts in the right polarization and entire IF
10) LEFT_filt: total power in counts in the left polarization and BBC
11) RIGHT_filt: total power in counts in the right polarization and BBC
12) CAL: states if the noise generator is ON or OFF
13) Marker: progressive number of the measured TPI

In order to plot the recorded graph, please click the PLOT button and you will see a new window where you will be able to set the X and Y values of your new graph.

In order to do this, FIRST you have to select the data you want to plot (for example, AZ_Offset) and THEN the button of the graph axis you want to assign (for example X).

For example, if you want to plot a Cross Scan graph, you can assign AZ_Offset to X and LEFT_filt to Y. Then please press PLOT to create the graph.

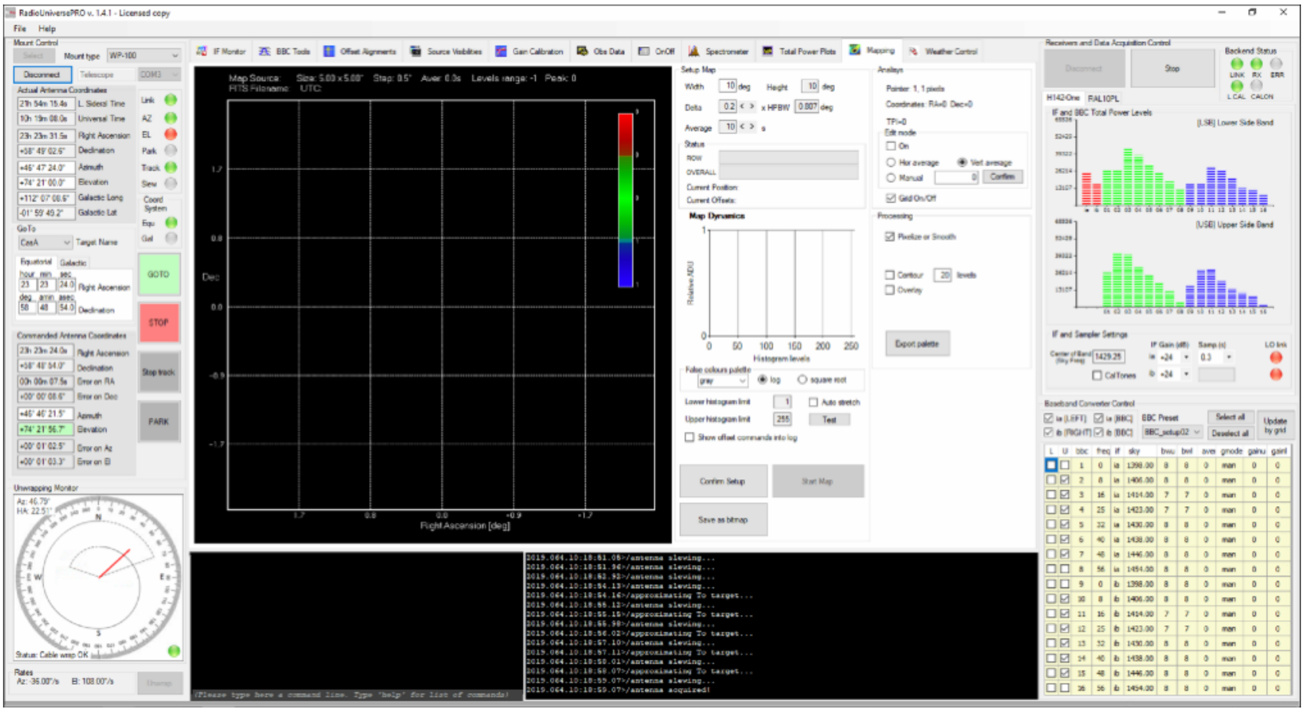




Mapping

Mapping tool lets you scan a sky area and convert it into an image map. Select the Mapping tab to reveal Mapping options in the Setup Map area. You can set:

- Width / Height: the side of the Map you want to create.
- Delta: a moltiplicative factor of the HPBW of your radio telescope. This is the distance from pixel to pixel of the map RadioUniversePRO will create. Reduce the value to reduce the antenna movement from pixel to pixel, both in Azimuth and Elevation.
- Average: number of seconds the antenna has to track every pixel and average values (the higher the value, the higher will be the S/N ratio).

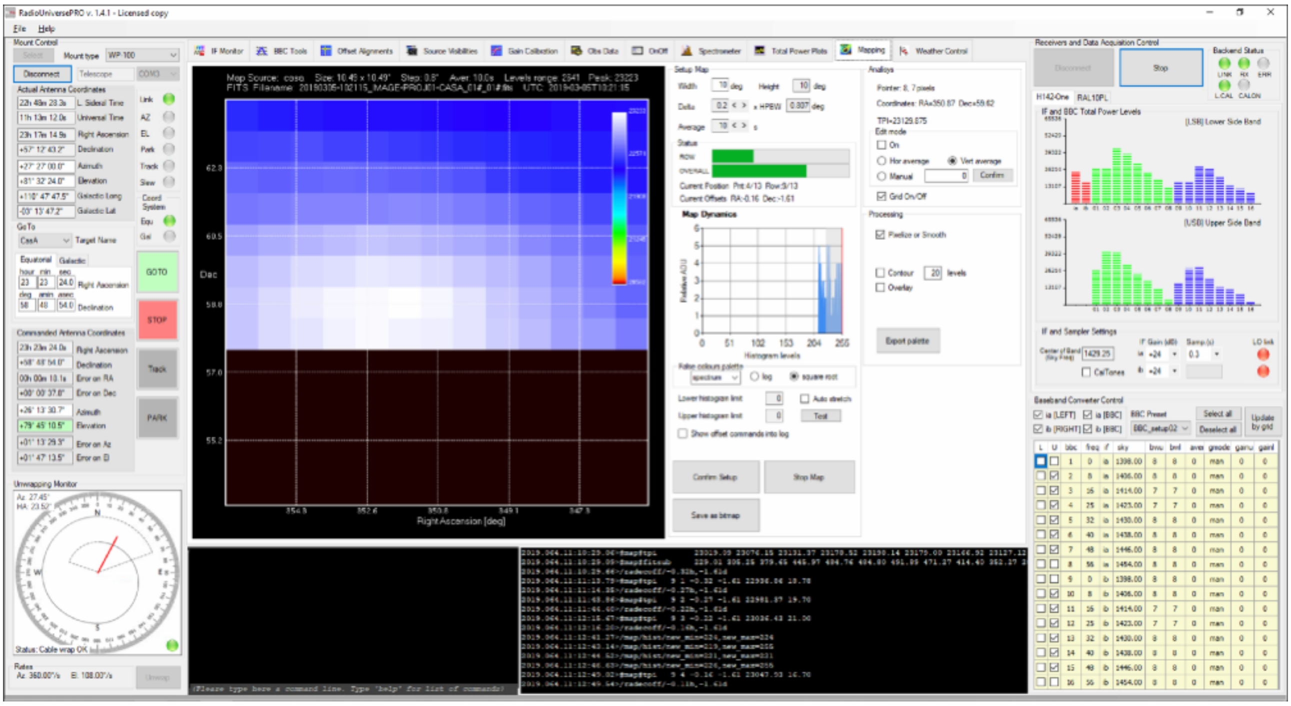


Below the Setup Map area you can find:

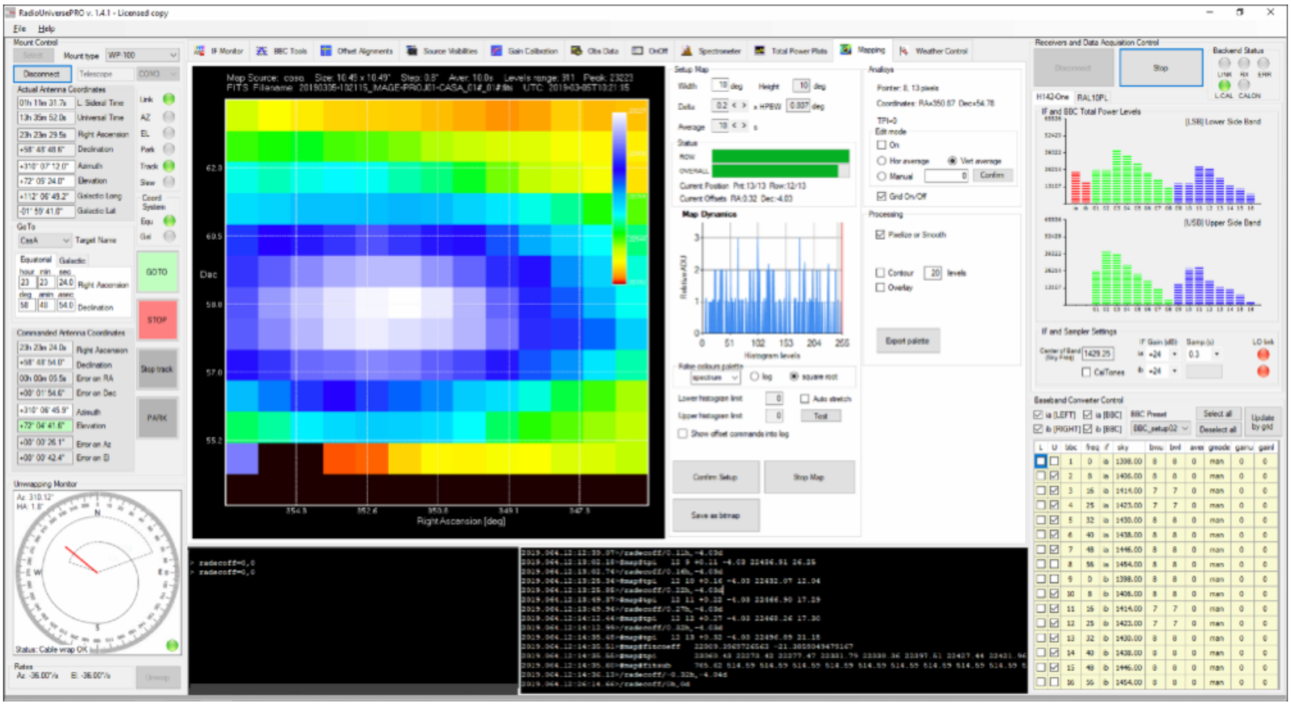
- Status: here you will see the progression of every ROW and the OVERALL status during capture.
- Map Dynamics: it shows the histogram of the map.
- False colours palette: here you can select the palette you want RadioUniversePRO to use to create the map and the "log" or "square root" methods.

When you set all the parameters, click on "Confirm Setup" to confirm and "Start Map" button to start map recording. You will see the radio telescope moving in different positions and the data will be progressive plotted in the map. The map shows the mapped object name, the map dimension and coordinates in Right Ascension and Declination.

TIP: when you want to record a radio map of the Sun you can set an Average value of 1 second. For all the other objects we suggest you to increase it to at least 10 seconds. Please note that a radio map, based on different settings you may choose, may take a long time to be recorded. This means that, before starting the record process for an entire radio map, you have to consider the actual radio telescope position and where the radio telescope will be after the time that is expected to record the entire map. We suggest you to avoid the radio telescope position to be too close to the horizon: it has to be at least 20 degrees from the horizon for the entire acquisition.



After every row acquisition, RadioUniversePRO will plot the row data and add to the map. During data acquisition, you can change the histogram visualization to better visualize the signal. In order to do so, move the mouse into the histogram window, click the left button of the mouse and, keeping the mouse clicked, drag the band up to the higher level you want to visualize. Release the mouse and the map will be updated.



When the Map is completed, you can press the “Save as bitmap” button to save it. You will also find the correspondent FITS file in the c:/programs/RadioUniversePRO folder. You also have many options to personalize the recorded map:

- Pixelize or Smooth
- Contour (you can set the number of levels in the proper field)
- Overlay

Manual controls for Input Window in RadioUniversePRO

The Input Window allow you to manually write commands that let the SPIDER radio telescope perform a specific action. Move the mouse on the lower line of the Inout Window area and type one of the commands described here, then press ENTER to execute:

1) In order to manually point the radio telescope to a specific direction you can type this command in Input Window area (then press keyboard ENTER button to launch the command):

source=azel,x,y

This command moves the radio telescope antenna to a coordinates defined position (x=azimuth angle, y=elevation angle) on the sky. For example, if you want to point the SPIDER radio telescope not to a defined radio object position (in this case you can make a double click on radio sources listed in Source Visibilities tab) but to the East and 45° from the horizon, you can simply write:

source=azel,90,45

2) By using the Total Power Plots tab you can measure in real time the variation of their radio flux data recorded by the SPIDER radio telescope by the time. When the receiver is powered on and connected, remember to select the proper BBC tool and then go to Total Power Plots and activate the view by clicking on "Tpi" option on the right column. Point one of the available radio sources (start with the Sun as example) and look at data in the graph. Then use this command:

azeloff=x,y

This command moves the radio telescope away from the radio sources by a defined angle in azimuth (x value) and elevation (y value). So if you write:

azeloff=5,5

The SPIDER radio telescope will move 5 degrees in azimuth and 5 degrees in elevation away from the radio source. Then click on the Total Power Plot tab and you will see the radio flux decreasing. In order to point the radio source, you can use the command

azeloff=0,0

This will reset the off angle values from the computed object position in the sky.

3) In order to park the radio telescope in stow position (before turning it off), please use this command:

source=azel,270,90

This will move the radio telescope in stow position (azimuth=270, elevation=90). When the radio telescope is in stow position (you can verify this by looking at it and looking at Actual Antenna Coordinates on the left column), please:

- disconnect the receiver by pressing Stop and Disconnect buttons on the right column
- Disconnect the mount by pressing the Disconnect button on the left column
- Exit from RadioUniversePRO software
- Turn off your computer and screen with the remote

4) In order to record a Cross Scan of an object, after having the SPIDER radio telescope pointed at a radio source (for example by using the Source Visibilities tab), you can also use this command:

crossscan=hor,5,1,2,if,1

The syntax of the command is

crossscan=[subscanFrame],[span],[step],[average],[devices],[save]

there:

subscanFrame = hor,equ,gal

span = the number in degrees that defines the length of the cross scan

step = the number in degrees that defines distance between two points in the cross scan

average = the number in seconds that defines how many seconds of integration are set for every cross scan point

devices = this can be “if” if you want to use the entire “if” for the totalpower or “bbcs” to use the bbc based on the actual setting of the filters

save = number that can be 1 or 0 and defines if you want to save cross scan data in FITS format

SPIDER 300A and 500A environmental conditions

The SPIDER 300A and 500A is designed for mount and antenna installation outside. The environmental conditions to be met for proper operation are:

	Condition
Temperature	from -10 °C / 14 °F to +50 °C / 122 °F
Maximum wind speed during use	50 Km/h (30 mph)
Maximum wind speed in stow position	150 Km/h (90 mph)

The H142-One receiver is installed inside, in a control room specially protected by environmental phenomena (rain, snow, hail). It is preferable to install the receiver in a room with temperature as constant as possible.

Extended temperature version: For operation in extreme temperature conditions (lower or higher than the above table condition), the user has to park the radio telescope in Stow Position and turn off the power.

Wind speed is one of the most important parameters to consider for the safe use of the radio telescope installed outside. SPIDER 300A or 500A radio telescopes are safe for installation in external environments and do not require coverage. For safe use, you must:

1) When not in use, the SPIDER 300A or 500A radio telescope are to be placed in the stow position with the antenna pointing to the vertical (Zenith). In this way the antenna offers the minimum wind resistance.

2) When in use, the SPIDER 300A or 500A radio telescope are designed for operations with winds up to 50 km/h (30 mph). When the wind speed is higher, place the SPIDER 300A or 500A radio telescope in Stow Position. The UltraSonic Wind Sensor for SPIDER radio telescopes (optional) allows you to constantly monitor the wind, using the RadioUniversePRO software provided with the radio telescope. This will automatically move the SPIDER 300A or 500A to stow position when the wind exceeds the 50 Km/h (30 Mph) safety threshold.

Electrical system

The instructions contained in the "Electrical System" section must be supplied to a Company specializing in electrical installations. These instructions are generally valid worldwide but may vary depending on the Country of installation of SPIDER, depending on the prescriptions of the electrical systems used in their respective Countries.

Power: the receiver and the power supply box are equipped with a power supply unit that must be connected to a power socket with the following characteristics:
- Worldwide (excluding USA and Canada): 230 V with Schuko socket or according to Country regulations. The Schuko connector is included in the SPIDER radio telescope.
- For USA and Canada: 115 V with AC outlet according to US standards.

Connection values: voltage variations over and above the normal levels compromise the proper operation of the SPIDER radio telescope. A voltage stabilizer should be used.

	SPIDER 300A	SPIDER 500A
Rated voltage at 50 Hz ± 1 %	230V	230V
Power absorption	500W	1600W

UPS. In order to avoid problems in case of sudden electricity cut off (that could stop the antenna in an unwanted position), we suggest you to use a UPS unit with the following advices:
- for the sizing of the UPS, the behavior in case of short-circuit and overload of the UPS system should also be considered.
- Overload capacity of the UPS system: ≥200% for 0.5 s.

Note: Sizing of the UPS must be absolutely defined by the supplier of the UPS!

Lightning protection. Due to the possible emission of radio waves and therefore the creation of strong interference, a lightning arrestor system can not be installed on the SPIDER radio telescope. The lightning protection system therefore should provide:

- Passive protection on the antenna + mount structure: the highest point of the radio telescope when positioned in stow position is the one that covers the 2 LNAs and it's made of non-metallic material. In the event of a lightning striking the SPIDER radio telescope, the electric current could freely flow through the entire structure and discharged to terrain. **SPIDER is provided with a special connection point on the column to be connected to the ground sink that the customer has to put in place according to the regulations in force in the Country of installation of the radio telescope.**
- Passive receiver protection: to prevent electric current from electrocution or electrostatic currents reaching to the receiver located in the radio telescope control room, **the SPIDER is equipped with special electrostatic voltage dischargers to be installed before the receiver: equipped with a special socket, it must be connected to the ground system, allowing the discharge of any electric current generated by the lightning strike before it reaches the receiver.**

SPIDER 300A and 500A power and data cables

The SPIDER 300A and 500A radio telescopes are made of elements to be installed outside (antenna and mount) and others to be installed inside (receiver and and control computers). All elements must be properly powered and must be connected to each other with special data cables provided with the radio telescope. In particular:

- ***Power***

The power supply of the external elements (antenna and mount) is made through the special cable that connects the mount to the control box to be placed in the control room and therefore no other external power point is required. The power supply must be provided starting from the control room where the 2 elements must be powered, connecting each one to a power socket.

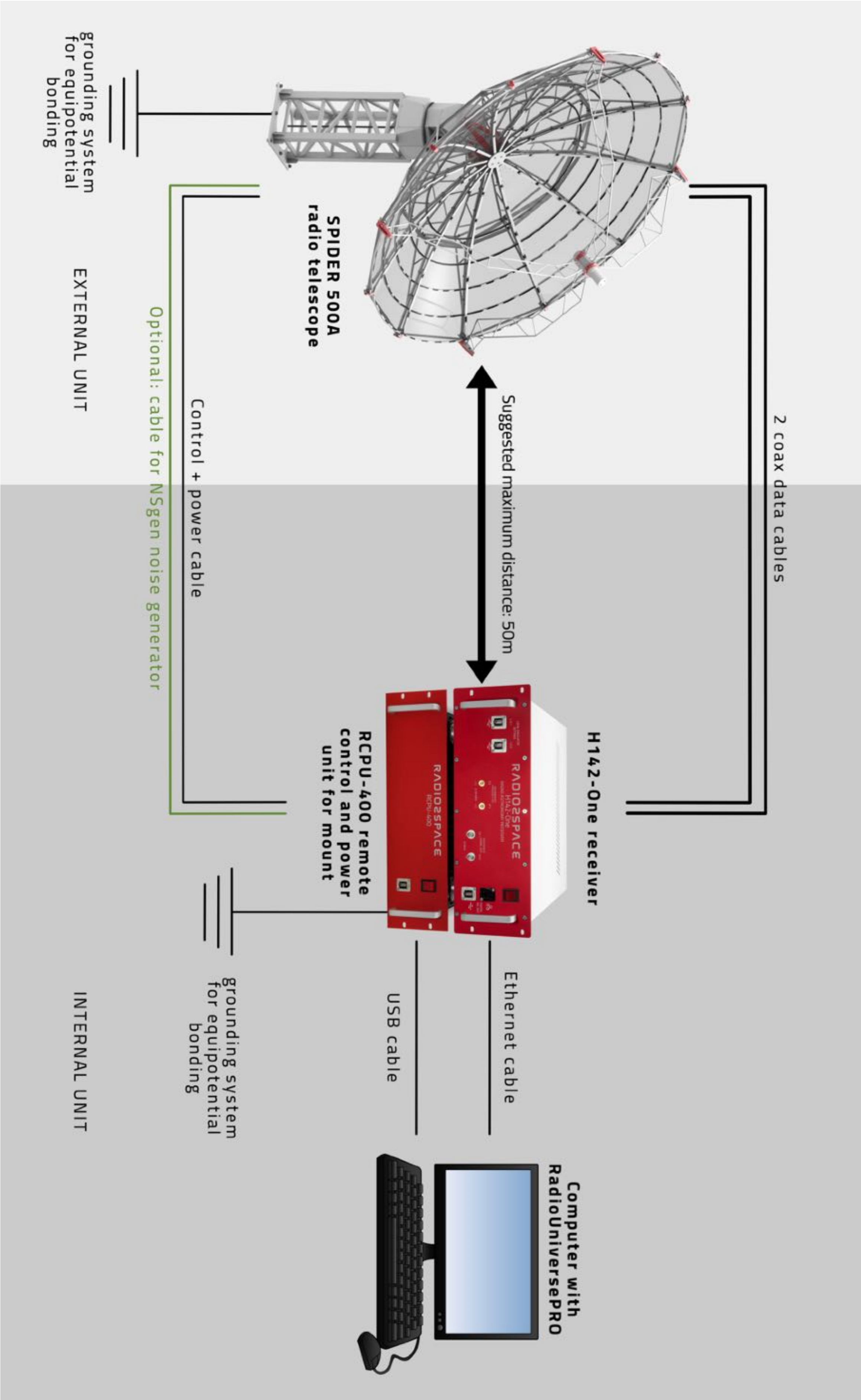
- ***Data and RF cables***

Data cables connect the external elements to the internal ones with a total of 3 cables. The 2 LNAs installed on the antenna are connected with 2 coaxial cables specially dimensioned to avoid data loss for distances (between external radio telescope and internal receiver) up to max 50 meters. Distances up to 100 meters are possible but this will introduce gain loss and we strongly suggest lower distances. To allow proper antenna handling, each coaxial cable is made up of 2 parts:

1) n.2 Ecoflex 10 coaxial cables that connect LNAs to the cables in the channel

2) n.2 Ecoflex 10 coaxial cables installed in the channel (as indicated by the Radio2Space project) and that connect the radio telescope to the receiver RF inputs placed in the control room at a maximum distance of 50 meters.

The mount is connected to the control computer via the same cable that also provides power to the motors (the cable has a multi-wire braided cable specially designed to avoid interference). The cable must be inserted into the channel connecting the outside radio telescope to the internal control room (along with the LNA connection cables). Inside the control room, the cable has a dedicated control box to be placed next to the receiver and connected to a control PC (where RadioUniversePRO software is installed) with a USB port.



grounding system
for equipotential
bonding
SPIDER 500A
radio telescope
EXTERNAL UNIT
Optional: cable for NSgen noise generator
Control + power cable
Suggested maximum distance: 50m
2 coax data cables
H142-One receiver
RCPU-400 remote
control and power
unit for mount
grounding system
for equipotential
bonding
USB cable
Ethernet cable
INTERNAL UNIT
Computer with
RadioUniversePRO

SPIDER radio telescopes maintenance

Since SPIDER radio telescopes are exposed to environmental phenomena that are variable depending on the geographic location where the radio telescope is installed, the following information must be followed:

- **Mechanical periodic control**: the user is required to periodically (we advice to make a check every month or every strong environmental effect like strong winds or sand storm) check the locking status of all the screws that are used in the antenna and mount. If necessary the user is requested is tighten the screws.

- **Grease change**: SPIDER 500A mounts comes with internal grease. The user is requested to periodically (every 3 months) check for the grease status and, at least every 6 months, change the grease with the grease given together with the radio telescope. This task can be easily accomplished by opening the proper windows in the mount head that give access to the mount worm gears, both in azimuth and elevation. Warning: the grease change operation has to be performed when the radio telescope is in Stow Position and powered off. Note: the grease change is not requested for the SPIDER 300A and 230C.

- **Clear from dust and water**: the user is requested to periodically (at least every 1 month or after every strong weather phenomena like a sand storm) check for the presence of dust and/or water in the mount head. If the user detect dust and/or water, the user is requested to clear it with the proper tools. Warning: this operation has to be performed when the radio telescope is in Stow Position and powered off.

- **Animals**: check that no animal may interfere with radio telescope operation. For example (but not restricted to this) birds could be on the antenna (especially when parked in Stow Position) or animals could byte the cables. The user is requested to check that no animal create problems to the radio telescope and, if a problem is detected, avoid the use of the radio telescope until the problem is fixed.

- **Snow**: when the antenna is in a parking position (Stow Position), it could load a lot of snow that can be accumulated on the parabolic antenna. In this case, the user is required to check the snow accumulation status and periodically point the antenna towards the horizon to favor the snow fall down from the parabolic antenna. If necessary the user will have to act manually to remove the snow but great attention must be applied not to damage the radio telescope, especially the antenna.

- **Ice**: in very cold conditions an important layer of ice may be formed on the parabolic antenna and/or on the mount. In normal cases this does not preclude the operation of the radio telescope but, if stored in large quantities, it may interfere with the handling of the antenna. In these cases the user is required to remove the ice before using the radio telescope, paying attention not to damage the radio telescope during the ice removal process. If it is not possible to remove the ice and it blocks (or tends to block) the movement of the radio telescope, the user shall favor the melting of the ice (eg using an air dryer or waiting for the solar irradiation melt down the ice layer).

- **Hailstorm**: in case of hail of significant size, the parabolic antenna could be damaged by the hail itself which could ruin the parabolic shape of the mesh until it buckles in case of very large hail. If the atmospheric conditions do permit (only if the wind is very low), the user can point the radio telescope antenna toward the horizon to reduce the risk of the antenna being hit by hail. The radio telescope has been designed with the detachable dish parts and, in case of severe damage, it is possible to replace the parabolic antenna (or part of it).

- **Intense hot/cold**: the radio telescope is designed for operation at temperatures between -10°C/14°F and +50°C/122°F. If you experience temperatures outside of this range, you are required to avoid using the radio telescope until conditions return to the correct range.

Warning: If hail (or other atmospheric phenomena) cause deformation or breakage of the parabolic antenna (or part of it), the user may order parabolic antenna (or part of it) replacement part.

Note: The meteorological events described in this paragraph are to be considered as exceptions, so any damage that may result from these phenomena is not covered by the product warranty, even though they occur during the warranty coverage period.

Troubleshooting

In order to check that the radio telescope correctly works, you can follow this very simple procedure:

1) ***check mount functionality***: during daytime, please point the Sun and, when the pointing is confirmed, check for the shadow of the feed.

   I. if you see the shadow in the center of the parabolic antenna (or very close to it): it means that the SPIDER's mount is working properly, the antenna is moving as requested and you may start an "Offset Alignment" procedure in order to even improve antenna alignment on the sky.

   II. if you see the shadow not in the center of the antenna: the mount is working but it needs to be aligned to the sky. First of all please check that all the data in "Obs Data" tab of RadioUniversePRO are correct. Then check the time and verify that it corresponds to the clock of your computer where you installed RadioUniversePRO software. If you find a setting error, fix it, disconnect from the mount, restart RadioUniversePRO, make a goto to the Sun (now the shadow should be closer to the center) and proceed with an "Offset Alignment" again.

   III. if you see the shadow far from the antenna (it means that the radio telescope is pointing a sky position far away from the Sun or it's not moving at all), first of all please check that the RCPU unit is tuned on and that the mount is correctly connected to the software as described in the installation manual. If it's correctly connected to the control computer, you have to read the mount position under "Actual Antenna Coordinate" left column in RadioUniversePRO. If you're reading all 00000000 values, it means that the mount is not connected to the control computer with RadioUniversePRO software. In this case please check all the connection cables (from the radio telescope mount to the RCPU unit), check that you selected the correct COM port number and the correct mount model in the upper part of the left column of RadioUniversePRO.

2) ***check receiver functionality***: if you're sure that the mount is working and it's aligned to the sky, during daytime please point the Sun and, when the pointing is confirmed, please check the signal in Power Spectrum, for both left and right polarizations. The Sun signal is so strong that it has to be visible in realtime even with very low digital gains (for example +6dB) set in the right column of RadioUniversePRO. In order to check it, with the radio telescope on and connected, please select the Sun in the sources lists and press the GOTO button. You should see the signal increase both in the Power Spectrum and in the FFT windows. When the source is acquired, you can test the signal by using the command azeloff=5,5 that will move the radio telescope away from the Sun by 5 degrees both in azimuth and in elevation and you should see a decrease of the signal. Then use the command azeloff=0,0 and this will point the radio telescope again to the Sun and you will see again the increase of the signal in both the polarizations. If you see this, everything is fine and your radio telescope is working perfectly. If you can't see signal increase (and you're sure your radio telescope is actually pointing at the Sun), please follow this guide in order to check all the parts of the receiving system of your SPIDER radio telescope:

   I. *Check the LNA units*: if you can't see signal in one polarization but you can see it in the other polarization, the first thing to check are the LNA units. Please disconnect receiver and mount from the RadioUniversePRO software, turn off all the power units and then invert RF cables in the rear ports of the receiver. Then turn on again all the power units and connect to the RadioUniversePRO software. Repeat the pointing to the Sun and check the power spectrum and FFT windows. If now you see that the signal is coming from the other polarization (and it's not coming from the polarization that before it was working), it means that one of the LNA is not working. You need to check cables connections to the LNA in the radio telescope (you need to move the radio telescope as close as possible to the horizon with the command source=azel, 270,15 then disconnect from the RadioUniversePRO software, turn off all the power units, open the cap the protects the LNA units on top of the feed and check that the cables are perfectly connected to the LNA units). If the cables are correctly connected to the LNA you can also remove the LNA units from the feed and directly connect the RF cables to the feed. Then turn on again all the power units of the radio telescope, connect to the RadioUniversePRO software and point the Sun. Without the LNA units the signal will be a lower but it has still to be visible (you can increase the digital gain in right column of RadioUniversePRO software). If now you see the signal, it means that everything is working except for the LNA unit (one or both) and you have to ship us back to fix the LNA (one or both, based on what unit you see it's not working). If you don't see the signal, please proceed to the next step and check cables.

II. *Check the RF cables*: every SPIDER radio telescope uses N type connectors for the RF cables, both male and female types. Male connectors are used for receiver connection, in the radio telescope pier and for LNA connection. Female connectors are used in the pier. In order to verify the cables, you have to check that the connector is properly connected to the cable itself since, for example during the insertion in the pipe connecting the radio telescope antenna to the control room, it could be disconnected.

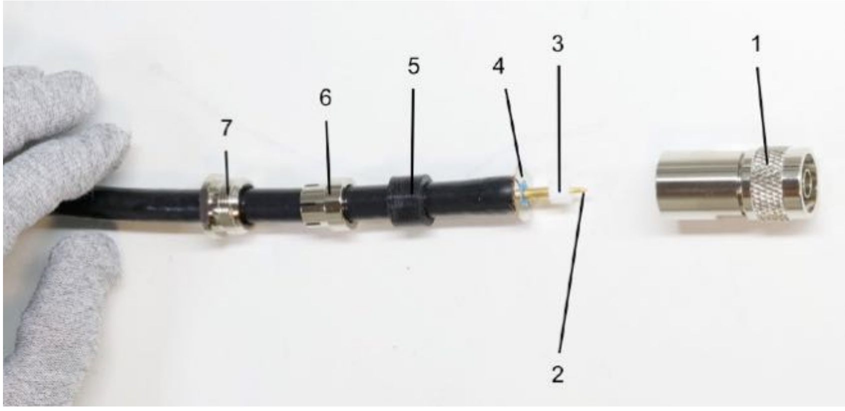
Female

Male

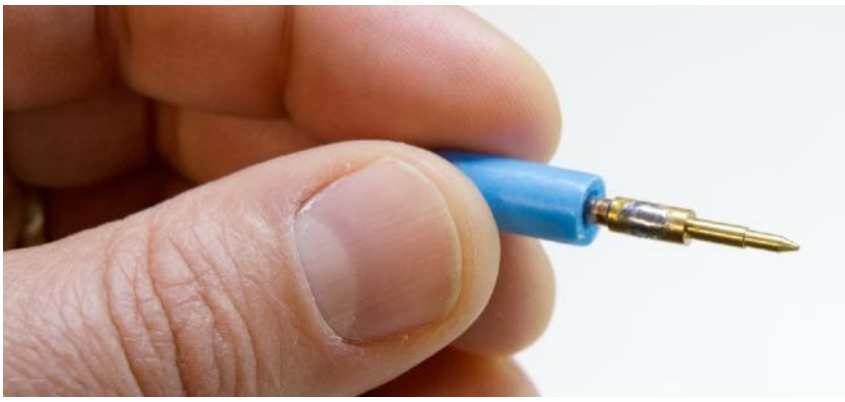
Check male connector: by using a 17 wrench, unscrew the rear cap of the connector.

Then remove the rear element

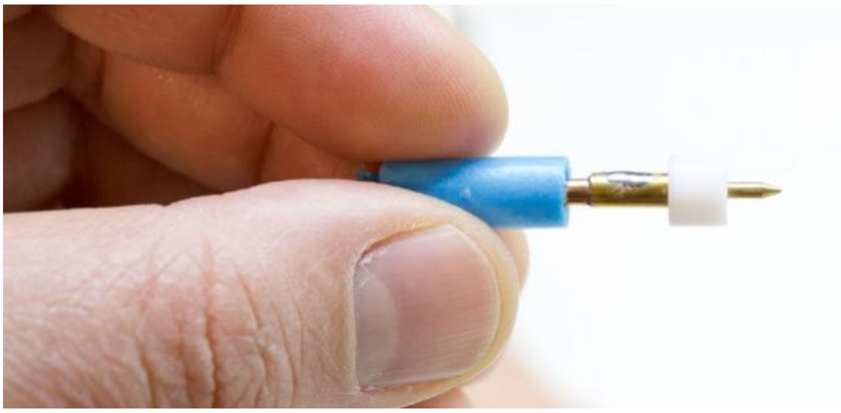
and open the connector as shown in the image below.



Check that the element 4 is clean, without residues of metallic shield (copper color) or foil (the copper sheet more internal to the metallic shield, adhering to the internal insulating material). There must be no traces of moisture or water, or any other element that could lead to a short. Check that the metallic shield does not directly touch the element 2. By using a magnifying glass, look at the element 2 and the relative soldering that fixes it on the central conductor of the RF cable.

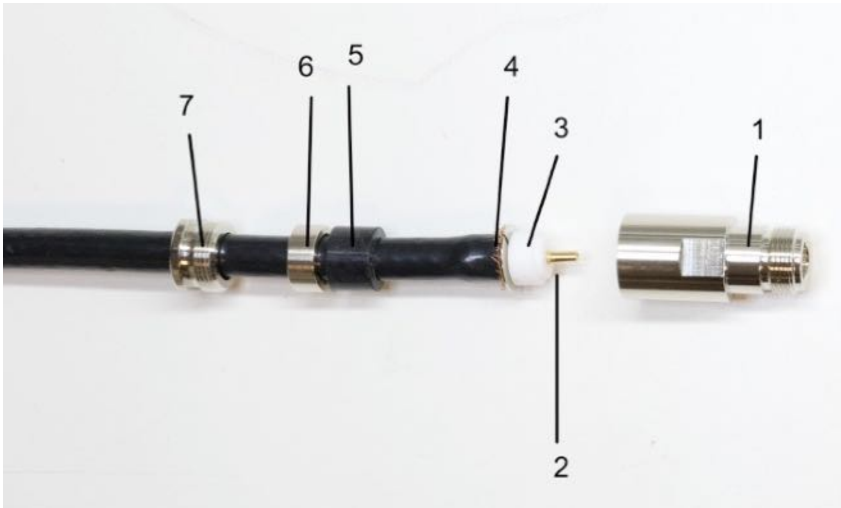


By applying a slight effort, check that the tip is fixed with respect to the central conductor (for example, holding it between two fingers and checking that it does not move). If everything is ok, insert again the element 3 on the tip (element 2) and close all the elements making sure that the element 2 is in the central position with respect to the terminal N.



Check female connector: by using a 17 wrench, unscrew the rear cap of the connector.

Then remove the rear element and open the connector as shown in the image below.



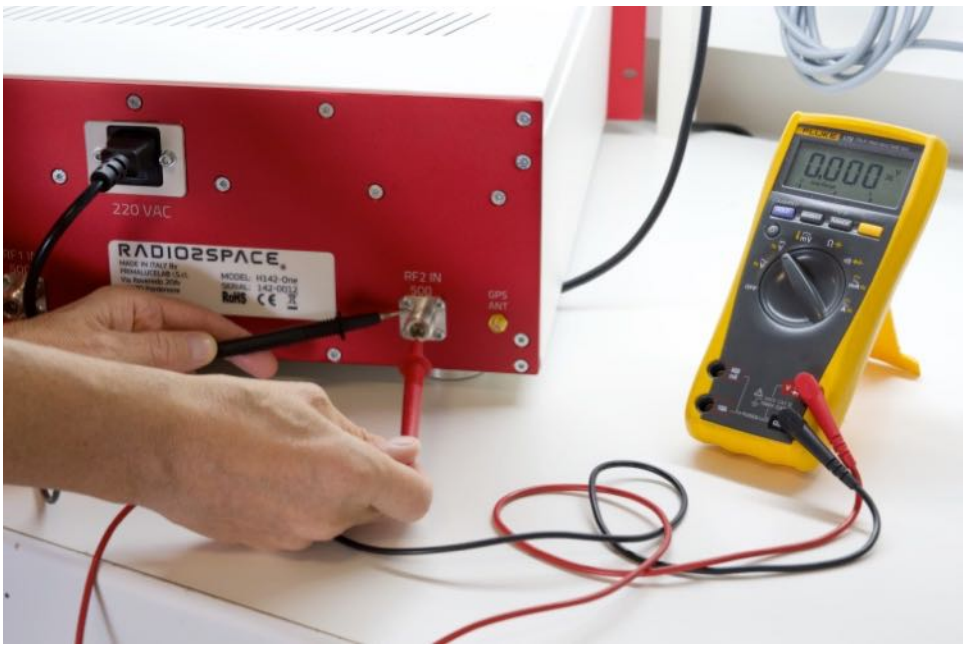
Check that the element 4 is clean, without residues of metallic shield (copper color) or foil (the copper sheet more internal to the metallic shield, adhering to the internal insulating material). There must be no traces of moisture or water, or any other element that could lead to a short. Check that the metallic shield does not directly touch the element 2.

By using a magnifying glass, look at the element 2 and the relative soldering that fixes it on the central conductor of the RF cable. By applying a slight effort, check that the tip is fixed with respect to the central conductor (for example, holding it between two fingers and checking that it does not move).

If everything is ok, insert again the element 3 on the tip (element 2) and close all the elements making sure that the element 2 is in the central position with respect to the terminal N.

After having checked that the connectors are properly set, by using a multimeter, check that there is not a short circuit between the tip (element 2) and the outside part of the N connector, for both male and female connectors.

III. *Check the receiver*: if you checked that LNA units and RF cables are ok but you still have one or two polarizations not showing the signal increase when you point the Sun, now you have to check the receiver. Having the receiver turned off, disconnect the RF cables from the back side of the receiver and then turn it on again. By using a tester, please check that the tension from the central pin to the external part of the connector is around 12V, for both the RF1 and RF2 ports. If you read around 12V, the receiver sends correctly the power to the LNA units.



If one of the two RF1 or RF2 channels (or both) doesn't provide power, you have to send us back the receiver for assistance. If both the channels provide power, proceed to the next step.

By using an oscilloscope connected to the 2 IF exit on the front panel of the receiver, please check you can read an analog signal.

If you see a signal it means that the receiver is sending data to the integrated spectrometer module and, but you can't see the signal in RadioUniversePRO, it means that the spectrometer module is not working. If you can't read any signal, it means that the internal analog receiver is not working. In both cases you have to send us back the receiver for assistance.

Note: when we build the every radio telescope, every component is tested in our laboratories before shipment and this is important because we know that everything is working before it is boxed and shipment. The radio telescope is composed by many electronic elements and sometimes something may fail. In this case, following this guide you can easily identify the part that is not perfectly working and contact us (info@radio2space.com or +39/0434/1696106) to get support.

Since the SPIDER radio telescope is composed by some elements that are installed outside, in order to prevent electronic problems it's mandatory the customer install proper grounding connections (for both the external antenna unit and the receiver in the control room) BEFORE the use of the radio telescope.